1)

The point on the x-axis closest to $(2,5,-3)$ is $(2,0,0)$

Distance $= \sqrt{(2-2)^2+(5-0)^2+(-3-0)^2} = \sqrt{34}$ units

Alternative Method:

Let A be the point with coordinates $(2,5,-3)$

The x-axis has equation $\mathbf{r} = \begin{pmatrix}0\\0\\0\end{pmatrix} + \lambda\begin{pmatrix}1\\0\\0\end{pmatrix}$

Let P be a point on the x-axis, so $\overrightarrow{OP} = \begin{pmatrix}\lambda\\0\\0\end{pmatrix}$

$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix}\lambda\\0\\0\end{pmatrix} - \begin{pmatrix}2\\5\\-3\end{pmatrix} = \begin{pmatrix}\lambda-2\\-5\\3\end{pmatrix}$

$\overrightarrow{AP} \cdot \begin{pmatrix}1\\0\\0\end{pmatrix} = 0 \Rightarrow \begin{pmatrix}\lambda-2\\-5\\3\end{pmatrix} \cdot \begin{pmatrix}1\\0\\0\end{pmatrix} = 0$

$\Rightarrow (\lambda-2)\times 1 + (-5)\times 0 + 3\times 0 = 0$

$\Rightarrow \lambda - 2 = 0$

$\lambda = 2$

So $\overrightarrow{AP} = \begin{pmatrix}2-2\\-5\\3\end{pmatrix} = \begin{pmatrix}0\\-5\\3\end{pmatrix}$

So $\left|\overrightarrow{AP}\right| = \sqrt{0^2+(-5)^2+3^2} = \sqrt{34}$ units

Alternative Method (using the Vector Product):

Let $\mathbf{p} = \begin{pmatrix}2\\5\\-3\end{pmatrix}$

Let A be a known point on the line so that its position vector is $\mathbf{a} = \begin{pmatrix}0\\0\\0\end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix}1\\0\\0\end{pmatrix}$

Then $h = \frac{|\mathbf{d}\times(\mathbf{p}-\mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix}1\\0\\0\end{pmatrix}\times\left(\begin{pmatrix}2\\5\\-3\end{pmatrix}-\begin{pmatrix}0\\0\\0\end{pmatrix}\right)\right|}{\left|\begin{pmatrix}1\\0\\0\end{pmatrix}\right|} = \frac{\left|\begin{pmatrix}1\\0\\0\end{pmatrix}\times\begin{pmatrix}2\\5\\-3\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\0\\0\end{pmatrix}\right|}$

$\begin{pmatrix}1\\0\\0\end{pmatrix}\times\begin{pmatrix}2\\5\\-3\end{pmatrix} = \begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&0&0\\2&5&-3\end{vmatrix}$

$= (0 \times (-3) - 0 \times 5)\mathbf{i} - (1 \times (-3) - 0 \times 2)\mathbf{j} + (1 \times 5 - 0 \times 2)\mathbf{k}$

$= 3\mathbf{j} + 5\mathbf{k}$

$\therefore h = \frac{\left|\begin{pmatrix}0\\3\\5\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\0\\0\end{pmatrix}\right|} = \frac{\sqrt{0^2+3^2+5^2}}{\sqrt{1^2+0^2+0^2}} = \frac{\sqrt{34}}{\sqrt{1}} = \sqrt{34}$ units

2)

The point on the z-axis closest to $(1, 8, -9)$ is $(0, 0, -9)$

Distance $= \sqrt{(1-0)^2 + (8-0)^2 + \big(-9-(-9)\big)^2} = \sqrt{65}$ units

Alternative Method:

Let A be the point with coordinates $(1, 8, -9)$

The z-axis has equation $\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

Let P be a point on the x-axis, so $\overrightarrow{OP} = \begin{pmatrix} 0 \\ 0 \\ \lambda \end{pmatrix}$

$$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} 0 \\ 0 \\ \lambda \end{pmatrix} - \begin{pmatrix} 1 \\ 8 \\ -9 \end{pmatrix} = \begin{pmatrix} -1 \\ -8 \\ \lambda + 9 \end{pmatrix}$$

$$\overrightarrow{AP} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} -1 \\ -8 \\ \lambda + 9 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0$$

$$\Rightarrow (-1) \times 0 + (-8) \times 0 + (\lambda + 9) \times 1 = 0$$

$$\Rightarrow \lambda + 9 = 0$$

$$\Rightarrow \lambda = -9$$

So $\overrightarrow{AP} = \begin{pmatrix} -1 \\ -8 \\ -9+9 \end{pmatrix} = \begin{pmatrix} -1 \\ -8 \\ 0 \end{pmatrix}$

So $\left|\overrightarrow{AP}\right| = \sqrt{(-1)^2 + (-8)^2 + 0^2} = \sqrt{65}$ units

Alternative Method (using the Vector Product):

Let $\mathbf{p} = \begin{pmatrix} 1 \\ 8 \\ -9 \end{pmatrix}$

Let A be a known point on the line so that its position vector is $\mathbf{a} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

Then $h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \times \left(\begin{pmatrix} 1 \\ 8 \\ -9 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right)\right|}{\left|\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right|} = \frac{\left|\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 8 \\ -9 \end{pmatrix}\right|}{\left|\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right|}$

$$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 8 \\ -9 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 1 \\ 1 & 8 & -9 \end{vmatrix}$$

$= (0 \times (-9) - 1 \times 8)\mathbf{i} - (0 \times (-9) - 1 \times 1)\mathbf{j} + (0 \times 8 - 0 \times 1)\mathbf{k}$

$= -8\mathbf{i} + \mathbf{j}$

$\therefore h = \frac{\left|\begin{pmatrix}-8\\1\\0\end{pmatrix}\right|}{\left|\begin{pmatrix}0\\0\\1\end{pmatrix}\right|} = \frac{\sqrt{(-8)^2+1^2+0^2}}{\sqrt{0^2+0^2+1^2}} = \frac{\sqrt{65}}{\sqrt{1}} = \sqrt{65}$ units

3)

$\left|\overrightarrow{OP}\right| = \sqrt{6^2 + 12^2 + (-5)^2} = \sqrt{205}$ units

4)

Let P be a point on the line, so $\overrightarrow{OP} = \begin{pmatrix} 1+2\lambda \\ 3+\lambda \\ 6+5\lambda \end{pmatrix}$

$\overrightarrow{OP} \cdot \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} 1+2\lambda \\ 3+\lambda \\ 6+5\lambda \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} = 0$

$\Rightarrow (1+2\lambda) \times 2 + (3+\lambda) \times 1 + (6+5\lambda) \times 5 = 0$

$\Rightarrow 2 + 4\lambda + 3 + \lambda + 30 + 25\lambda = 0$

$\Rightarrow 30\lambda + 35 = 0$

$\lambda = -\frac{7}{6}$

So $\overrightarrow{OP} = \begin{pmatrix} 1+2\left(-\frac{7}{6}\right) \\ 3+\left(-\frac{7}{6}\right) \\ 6+5\left(-\frac{7}{6}\right) \end{pmatrix} = \begin{pmatrix} -\frac{4}{3} \\ \frac{11}{6} \\ \frac{1}{6} \end{pmatrix}$

So $\left|\overrightarrow{OP}\right| = \sqrt{\left(-\frac{4}{3}\right)^2 + \left(\frac{11}{6}\right)^2 + \left(\frac{1}{6}\right)^2} = \frac{\sqrt{186}}{6}$ units

Alternative Method (using the Vector Product):

Let $\mathbf{p} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

Let A be a known point on the line so that its position vector is $\mathbf{a} = \begin{pmatrix} 1 \\ 3 \\ 6 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}$

Then $h = \frac{|\mathbf{d} \times (\mathbf{p}-\mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} \times \left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 3 \\ 6 \end{pmatrix}\right)\right|}{\left|\begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}\right|} = \frac{\left|\begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} \times \begin{pmatrix} -1 \\ -3 \\ -6 \end{pmatrix}\right|}{\left|\begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}\right|}$

$\begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} \times \begin{pmatrix} -1 \\ -3 \\ -6 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 5 \\ -1 & -3 & -6 \end{vmatrix}$

$= \big(1 \times (-6) - 5 \times (-3)\big)\mathbf{i} - \big(2 \times (-6) - 5 \times (-1)\big)\mathbf{j} + \big(2 \times (-3) - 1 \times (-1)\big)\mathbf{k}$

$= 9\mathbf{i} + 7\mathbf{j} - 5\mathbf{k}$

$\therefore h = \frac{\left|\begin{pmatrix} 9 \\ 7 \\ -5 \end{pmatrix}\right|}{\left|\begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}\right|} = \frac{\sqrt{9^2+7^2+(-5)^2}}{\sqrt{2^2+1^2+5^2}} = \frac{\sqrt{155}}{\sqrt{30}} = \frac{\sqrt{186}}{6}$ units

5)

Let A be the point with coordinates $(2, 1, 1)$

Let P be a point on the line, so $\overrightarrow{OP} = \begin{pmatrix} 5-2\lambda \\ 3+\lambda \\ -4+3\lambda \end{pmatrix}$

$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} 5-2\lambda \\ 3+\lambda \\ -4+3\lambda \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3-2\lambda \\ 2+\lambda \\ -5+3\lambda \end{pmatrix}$

$\overrightarrow{AP} \cdot \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} 3-2\lambda \\ 2+\lambda \\ -5+3\lambda \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} = 0$

$\Rightarrow (3-2\lambda) \times (-2) + (2+\lambda) \times 1 + (-5+3\lambda) \times 3 = 0$

$\Rightarrow -6 + 4\lambda + 2 + \lambda - 15 + 9\lambda = 0$

$\Rightarrow 14\lambda - 19 = 0$

$\lambda = \frac{19}{14}$

So $\overrightarrow{AP} = \begin{pmatrix} 3-2\left(\frac{19}{14}\right) \\ 2+\left(\frac{19}{14}\right) \\ -5+3\left(\frac{19}{14}\right) \end{pmatrix} = \begin{pmatrix} \frac{2}{7} \\ \frac{47}{14} \\ -\frac{13}{14} \end{pmatrix}$

So $\left|\overrightarrow{AP}\right| = \sqrt{\left(\frac{2}{7}\right)^2 + \left(\frac{47}{14}\right)^2 + \left(-\frac{13}{14}\right)^2} = \frac{3\sqrt{266}}{14}$ units

Alternative Method (using the Vector Product):

Let $\mathbf{p} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$

Let A be a known point on the line so that its position vector is $\mathbf{a} = \begin{pmatrix} 5 \\ 3 \\ -4 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix}$

Then $h = \frac{|\mathbf{d} \times (\mathbf{p}-\mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} \times \left(\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 5 \\ 3 \\ -4 \end{pmatrix}\right)\right|}{\left|\begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix}\right|} = \frac{\left|\begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} \times \begin{pmatrix} -3 \\ -2 \\ 5 \end{pmatrix}\right|}{\left|\begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix}\right|}$

$\begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} \times \begin{pmatrix} -3 \\ -2 \\ 5 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 1 & 3 \\ -3 & -2 & 5 \end{vmatrix}$

$= \big(1 \times 5 - 3 \times (-2)\big)\mathbf{i} - \big((-2) \times 5 - 3 \times (-3)\big)\mathbf{j} + \big((-2) \times (-2) - 1 \times (-3)\big)\mathbf{k}$

$= 11\mathbf{i} + \mathbf{j} + 7\mathbf{k}$

$$\therefore h = \frac{\left|\begin{pmatrix}11\\1\\7\end{pmatrix}\right|}{\left|\begin{pmatrix}-2\\1\\3\end{pmatrix}\right|} = \frac{\sqrt{11^2+1^2+7^2}}{\sqrt{(-2)^2+1^2+3^2}} = \frac{\sqrt{171}}{\sqrt{14}} = \frac{3\sqrt{266}}{14} \text{ units}$$

6)

Let A be the point with coordinates $(-4, 3, 8)$

The line has equation $\mathbf{r} = \begin{pmatrix}1\\0\\1\end{pmatrix} + \lambda\begin{pmatrix}1\\1\\0\end{pmatrix}$

Let P be a point on the line, so $\overrightarrow{OP} = \begin{pmatrix}1+\lambda\\ \lambda \\ 1\end{pmatrix}$

$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix}1+\lambda\\ \lambda \\ 1\end{pmatrix} - \begin{pmatrix}-4\\3\\8\end{pmatrix} = \begin{pmatrix}5+\lambda\\ \lambda-3 \\ -7\end{pmatrix}$

$\overrightarrow{AP} \cdot \begin{pmatrix}1\\1\\0\end{pmatrix} = 0 \Rightarrow \begin{pmatrix}5+\lambda\\ \lambda-3 \\ -7\end{pmatrix} \cdot \begin{pmatrix}1\\1\\0\end{pmatrix} = 0$

$\Rightarrow (5+\lambda) \times 1 + (\lambda - 3) \times 1 + (-7) \times 0 = 0$

$\Rightarrow 5 + \lambda + \lambda - 3 = 0$

$\Rightarrow 2\lambda + 2 = 0$

$\lambda = -1$

So $\overrightarrow{AP} = \begin{pmatrix}5-1\\ -1-3 \\ -7\end{pmatrix} = \begin{pmatrix}4\\ -4 \\ -7\end{pmatrix}$

So $\left|\overrightarrow{AP}\right| = \sqrt{4^2 + (-4)^2 + (-7)^2} = \sqrt{81} = 9$

So $r = 9$

Alternative Method (using the Vector Product):

Let $\mathbf{p} = \begin{pmatrix}-4\\3\\8\end{pmatrix}$

Let A be a known point on the line so that its position vector is $\mathbf{a} = \begin{pmatrix}1\\0\\1\end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix}1\\1\\0\end{pmatrix}$

Then $h = \frac{|\mathbf{d}\times(\mathbf{p}-\mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix}1\\1\\0\end{pmatrix}\times\left(\begin{pmatrix}-4\\3\\8\end{pmatrix}-\begin{pmatrix}1\\0\\1\end{pmatrix}\right)\right|}{\left|\begin{pmatrix}1\\1\\0\end{pmatrix}\right|} = \frac{\left|\begin{pmatrix}1\\1\\0\end{pmatrix}\times\begin{pmatrix}-5\\3\\7\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\1\\0\end{pmatrix}\right|}$

$\begin{pmatrix}1\\1\\0\end{pmatrix} \times \begin{pmatrix}-5\\3\\7\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ 1 & 1 & 0\\ -5 & 3 & 7\end{vmatrix}$

$= (1 \times 7 - 0 \times 3)\mathbf{i} - \big(1 \times 7 - 0 \times (-5)\big)\mathbf{j} + \big(1 \times 3 - 1 \times (-5)\big)\mathbf{k}$

$= 7\mathbf{i} - 7\mathbf{j} + 8\mathbf{k}$

$\therefore h = \frac{\left|\begin{pmatrix} 7 \\ -7 \\ 8 \end{pmatrix}\right|}{\left|\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}\right|} = \frac{\sqrt{7^2+(-7)^2+8^2}}{\sqrt{1^2+1^2+0^2}} = \frac{\sqrt{162}}{\sqrt{2}} = \sqrt{81} = 9$

So $r = 9$

7)

$\frac{x-1}{2} = y = z+2 \Rightarrow \frac{x-1}{2} = \frac{y}{1} = \frac{z+2}{1}$

Let l be the line $\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$

Let A be the point with coordinates $(1, 0, 1)$

Let P be a point on the line, so $\overrightarrow{OP} = \begin{pmatrix} 1+2\lambda \\ \lambda \\ -2+\lambda \end{pmatrix}$

$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} 1+2\lambda \\ \lambda \\ -2+\lambda \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2\lambda \\ \lambda \\ \lambda - 3 \end{pmatrix}$

$\overrightarrow{AP} \cdot \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} 2\lambda \\ \lambda \\ \lambda - 3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = 0$

$\Rightarrow 2\lambda \times 2 + \lambda \times 1 + (\lambda - 3) \times 1 = 0$

$\Rightarrow 4\lambda + \lambda + \lambda - 3 = 0$

$\Rightarrow 6\lambda - 3 = 0$

$\Rightarrow \lambda = \frac{1}{2}$

So $\overrightarrow{AP} = \begin{pmatrix} 2\left(\frac{1}{2}\right) \\ \frac{1}{2} \\ \left(\frac{1}{2}\right) - 3 \end{pmatrix} = \begin{pmatrix} 1 \\ \frac{1}{2} \\ -\frac{5}{2} \end{pmatrix}$

So $\left|\overrightarrow{AP}\right| = \sqrt{1^2 + \left(\frac{1}{2}\right)^2 + \left(-\frac{5}{2}\right)^2} = \sqrt{\frac{15}{2}} = \frac{\sqrt{30}}{2}$ units

Alternative Method (using the Vector Product):

$\frac{x-1}{2} = y = z+2 \Rightarrow \frac{x-1}{2} = \frac{y}{1} = \frac{z+2}{1}$

Let l be the line $\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$

Let $\mathbf{p} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

Let A be a known point on the line so that its position vector is $\mathbf{a} = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$

Then $h = \frac{|\mathbf{d}\times(\mathbf{p}-\mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix}2\\1\\1\end{pmatrix}\times\left(\begin{pmatrix}1\\0\\1\end{pmatrix}-\begin{pmatrix}1\\0\\-2\end{pmatrix}\right)\right|}{\left|\begin{pmatrix}2\\1\\1\end{pmatrix}\right|} = \frac{\left|\begin{pmatrix}2\\1\\1\end{pmatrix}\times\begin{pmatrix}0\\0\\3\end{pmatrix}\right|}{\left|\begin{pmatrix}2\\1\\1\end{pmatrix}\right|}$

$\begin{pmatrix}2\\1\\1\end{pmatrix}\times\begin{pmatrix}0\\0\\3\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\2 & 1 & 1\\0 & 0 & 3\end{vmatrix}$

$= (1\times 3 - 1\times 0)\mathbf{i} - (2\times 3 - 1\times 0)\mathbf{j} + (2\times 0 - 1\times 0)\mathbf{k}$

$= 3\mathbf{i} - 6\mathbf{j}$

$\therefore h = \frac{\left|\begin{pmatrix}3\\-6\\0\end{pmatrix}\right|}{\left|\begin{pmatrix}2\\1\\1\end{pmatrix}\right|} = \frac{\sqrt{3^2+(-6)^2+0^2}}{\sqrt{2^2+1^2+1^2}} = \frac{\sqrt{45}}{\sqrt{6}} = \frac{\sqrt{30}}{2}$ units

8)

$x = 2y = 3 - z \Rightarrow \frac{x}{1} = \frac{y}{\frac{1}{2}} = \frac{z-3}{-1}$

Let l be the line $\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ \frac{1}{2} \\ -1 \end{pmatrix}$ or $\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$

Let A be the point with coordinates $(6, 0, 4)$

Let P be a point on the line, so $\overrightarrow{OP} = \begin{pmatrix} 2\lambda \\ \lambda \\ 3 - 2\lambda \end{pmatrix}$

$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} 2\lambda \\ \lambda \\ 3 - 2\lambda \end{pmatrix} - \begin{pmatrix} 6 \\ 0 \\ 4 \end{pmatrix} = \begin{pmatrix} 2\lambda - 6 \\ \lambda \\ -1 - 2\lambda \end{pmatrix}$

$\overrightarrow{AP} \cdot \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} 2\lambda - 6 \\ \lambda \\ -1 - 2\lambda \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} = 0$

$\Rightarrow (2\lambda - 6) \times 2 + \lambda \times 1 + (-1 - 2\lambda) \times (-2) = 0$

$\Rightarrow 4\lambda - 12 + \lambda + 2 + 4\lambda = 0$

$\Rightarrow 9\lambda - 10 = 0$

$\lambda = \frac{10}{9}$

So $\overrightarrow{AP} = \begin{pmatrix} 2\left(\frac{10}{9}\right) - 6 \\ \frac{10}{9} \\ -1 - 2\left(\frac{10}{9}\right) \end{pmatrix} = \begin{pmatrix} -\frac{34}{9} \\ \frac{10}{9} \\ -\frac{29}{9} \end{pmatrix}$

So $\left|\overrightarrow{AP}\right| = \sqrt{\left(-\frac{34}{9}\right)^2 + \left(\frac{10}{9}\right)^2 + \left(-\frac{29}{9}\right)^2} = \frac{\sqrt{233}}{3}$ units

Alternative Method (using the Vector Product):

$x = 2y = 3 - z \Rightarrow \frac{x}{1} = \frac{y}{\frac{1}{2}} = \frac{z-3}{-1}$

Let l be the line $\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ \frac{1}{2} \\ -1 \end{pmatrix}$ or $\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$

Let $\mathbf{p} = \begin{pmatrix} 6 \\ 0 \\ 4 \end{pmatrix}$

Let A be a known point on the line so that its position vector is $\mathbf{a} = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$

Then $h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} \times \left(\begin{pmatrix} 6 \\ 0 \\ 4 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}\right)\right|}{\left|\begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}\right|} = \frac{\left|\begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} \times \begin{pmatrix} 6 \\ 0 \\ 1 \end{pmatrix}\right|}{\left|\begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}\right|}$

$$\begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} \times \begin{pmatrix} 6 \\ 0 \\ 1 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & -2 \\ 6 & 0 & 1 \end{vmatrix}$$

$$= (1 \times 1 - (-2) \times 0)\mathbf{i} - (2 \times 1 - (-2) \times 6)\mathbf{j} + (2 \times 0 - 1 \times 6)\mathbf{k}$$

$$= \mathbf{i} - 14\mathbf{j} - 6\mathbf{k}$$

$\therefore h = \frac{\left|\begin{pmatrix} 1 \\ -14 \\ -6 \end{pmatrix}\right|}{\left|\begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}\right|} = \frac{\sqrt{1^2 + (-14)^2 + (-6)^2}}{\sqrt{2^2 + 1^2 + (-2)^2}} = \frac{\sqrt{233}}{\sqrt{9}} = \frac{\sqrt{233}}{3}$ units

9) a)

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 9 \\ 12 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 \\ 13 \\ 3 \end{pmatrix}$

Let l be the line $\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 7 \\ 13 \\ 3 \end{pmatrix}$

$9 = 2 + 7\lambda \Rightarrow \lambda = 1$

$-22 = -1 + 13\lambda \Rightarrow \lambda = -\frac{21}{13}$

$0 = 1 + 3\lambda \Rightarrow \lambda = -\frac{1}{3}$

Inconsistent values of λ

$\therefore P(9, -22, 0)$ does not lie on the line

9) b)

Let Q be a point on the line, so $\overrightarrow{OQ} = \begin{pmatrix} 2+7\lambda \\ -1+13\lambda \\ 1+3\lambda \end{pmatrix}$

$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = \begin{pmatrix} 2+7\lambda \\ -1+13\lambda \\ 1+3\lambda \end{pmatrix} - \begin{pmatrix} 9 \\ -22 \\ 0 \end{pmatrix} = \begin{pmatrix} -7+7\lambda \\ 21+13\lambda \\ 1+3\lambda \end{pmatrix}$

$\overrightarrow{PQ} \cdot \begin{pmatrix} 7 \\ 13 \\ 3 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} -7+7\lambda \\ 21+13\lambda \\ 1+3\lambda \end{pmatrix} \cdot \begin{pmatrix} 7 \\ 13 \\ 3 \end{pmatrix} = 0$

$\Rightarrow (-7+7\lambda) \times 7 + (21+13\lambda) \times 13 + (1+3\lambda) \times 3 = 0$

$\Rightarrow -49 + 49\lambda + 273 + 169\lambda + 3 + 9\lambda = 0$

$\Rightarrow 227\lambda + 227 = 0$

$\lambda = -1$

So $\overrightarrow{OQ} = \begin{pmatrix} 2+7(-1) \\ -1+13(-1) \\ 1+3(-1) \end{pmatrix} = \begin{pmatrix} -5 \\ -14 \\ -2 \end{pmatrix}$

So the coordinates of Q are $(-5, -14, -2)$

9) c)

So $\overrightarrow{PQ} = \begin{pmatrix} -7+7(-1) \\ 21+13(-1) \\ 1+3(-1) \end{pmatrix} = \begin{pmatrix} -14 \\ 8 \\ -2 \end{pmatrix}$

So $\left|\overrightarrow{PQ}\right| = \sqrt{(-14)^2+8^2+(-2)^2} = 2\sqrt{66}$ units

Alternative Method (using the Vector Product):

Let $\mathbf{p} = \begin{pmatrix} 9 \\ -22 \\ 0 \end{pmatrix}$

Let A be a known point on the line so that its position vector is $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 7 \\ 13 \\ 3 \end{pmatrix}$

Then $h = \frac{|\mathbf{d}\times(\mathbf{p}-\mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix} 7 \\ 13 \\ 3 \end{pmatrix}\times\left(\begin{pmatrix} 9 \\ -22 \\ 0 \end{pmatrix}-\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}\right)\right|}{\left|\begin{pmatrix} 7 \\ 13 \\ 3 \end{pmatrix}\right|} = \frac{\left|\begin{pmatrix} 7 \\ 13 \\ 3 \end{pmatrix}\times\begin{pmatrix} 7 \\ -21 \\ -1 \end{pmatrix}\right|}{\left|\begin{pmatrix} 7 \\ 13 \\ 3 \end{pmatrix}\right|}$

$$\begin{pmatrix} 7 \\ 13 \\ 3 \end{pmatrix}\times\begin{pmatrix} 7 \\ -21 \\ -1 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 7 & 13 & 3 \\ 7 & -21 & -1 \end{vmatrix}$$

$$= \big(13\times(-1)-3\times(-21)\big)\mathbf{i} - (7\times(-1)-3\times 7)\mathbf{j} + (7\times(-21)-13\times 7)\mathbf{k}$$

$$= 50\mathbf{i}+28\mathbf{j}-238\mathbf{k}$$

$$\therefore h = \frac{\left|\begin{pmatrix} 50 \\ 28 \\ -238 \end{pmatrix}\right|}{\left|\begin{pmatrix} 7 \\ 13 \\ 3 \end{pmatrix}\right|} = \frac{\sqrt{50^2+28^2+(-238)^2}}{\sqrt{7^2+13^2+3^2}} = \frac{\sqrt{59928}}{\sqrt{227}} = 2\sqrt{66} \text{ units}$$

10)

Midpoint of A and B is $\left(\frac{3+5}{2}, \frac{-1+1}{2}, \frac{0+2}{2}\right) = (4, 0, 1)$

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 5 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$

The direction vector of the line, $\mathbf{d}$, must be perpendicular to the normal vector of the plane: $\mathbf{d} \cdot \begin{pmatrix} 2 \\ 7 \\ -9 \end{pmatrix} = 0$

And d must also be perpendicular to AB: $\mathbf{d} \cdot \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} = 0$

Let $\mathbf{d} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$

$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 7 \\ -9 \end{pmatrix} = 0 \Rightarrow 2a + 7b - 9c = 0$

$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} = 0 \Rightarrow 2a + 2b + 2c = 0$

Let $c = 1$, then the equations become:

$2a + 7b - 9 = 0 \Rightarrow 2a + 7b = 9$ (1)

$2a + 2b + 2 = 0 \Rightarrow a + b = -1$ (2)

Solve (1) and (2) to find $a = -\frac{16}{5}$ and $b = \frac{11}{5}$

Then the equation of the line is $\mathbf{r} = \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -\frac{16}{5} \\ \frac{11}{5} \\ 1 \end{pmatrix}$ or $\mathbf{r} = \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -16 \\ 11 \\ 5 \end{pmatrix}$

11) a)

Substitute $P(2, 2, 8)$ into the plane Π:

$3(2) - 8(2) + 2(8) - 6 = 6 - 16 + 16 - 6 = 0$ as required

Substitute $Q(10, 3, 0)$ into the plane Π:

$3(10) - 8(3) + 2(0) - 6 = 30 - 24 + 0 - 6 = 0$ as required

$\therefore$ both P and Q lie on the plane Π

11) b)

Midpoint of P and Q is $\left(\frac{2+10}{2}, \frac{2+3}{2}, \frac{8+0}{2}\right) = \left(6, \frac{5}{2}, 4\right)$

$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = \begin{pmatrix} 10 \\ 3 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ 2 \\ 8 \end{pmatrix} = \begin{pmatrix} 8 \\ 1 \\ -8 \end{pmatrix}$

The direction vector of the line, $\mathbf{d}$, must be perpendicular to the normal vector of the plane: $\mathbf{d} \cdot \begin{pmatrix} 3 \\ -8 \\ 2 \end{pmatrix} = 0$

And d must also be perpendicular to PQ: $\mathbf{d} \cdot \begin{pmatrix} 8 \\ 1 \\ -8 \end{pmatrix} = 0$

Let $\mathbf{d} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$

$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -8 \\ 2 \end{pmatrix} = 0 \Rightarrow 3a - 8b + 2c = 0$

$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} 8 \\ 1 \\ -8 \end{pmatrix} = 0 \Rightarrow 8a + b - 8c = 0$

Let $c = 1$, then the equations become:

$3a - 8b + 2 = 0 \Rightarrow 3a - 8b = -2$ (1)

$8a + b - 8 = 0 \Rightarrow 8a + b = 8$ (2)

Solve (1) and (2) to find $a = \frac{62}{67}$ and $b = \frac{40}{67}$

Then the equation of the line is $\mathbf{r} = \begin{pmatrix} 6 \\ \frac{5}{2} \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} \frac{62}{67} \\ \frac{40}{67} \\ 1 \end{pmatrix}$ or $\mathbf{r} = \begin{pmatrix} 6 \\ \frac{5}{2} \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 62 \\ 40 \\ 67 \end{pmatrix}$

11) c)

$$\overrightarrow{OR} = \begin{pmatrix} 6+62\lambda \\ \frac{5}{2}+40\lambda \\ 4+67\lambda \end{pmatrix}$$

$$\overrightarrow{PR} = \overrightarrow{OR} - \overrightarrow{OP} = \begin{pmatrix} 6+62\lambda \\ \frac{5}{2}+40\lambda \\ 4+67\lambda \end{pmatrix} - \begin{pmatrix} 2 \\ 2 \\ 8 \end{pmatrix} = \begin{pmatrix} 4+62\lambda \\ \frac{1}{2}+40\lambda \\ -4+67\lambda \end{pmatrix}$$

$$\left|\overrightarrow{PR}\right| = \frac{43}{\sqrt{6}} \Rightarrow \sqrt{(4+62\lambda)^2 + \left(\frac{1}{2}+40\lambda\right)^2 + (-4+67\lambda)^2} = \frac{43}{\sqrt{6}}$$

$$\Rightarrow (4+62\lambda)^2 + \left(\frac{1}{2}+40\lambda\right)^2 + (-4+67\lambda)^2 = \frac{1849}{6}$$

$$\Rightarrow 16 + 496\lambda + 3844\lambda^2 + \frac{1}{4} + 40\lambda + 1600\lambda^2 + 16 - 536\lambda + 4489\lambda^2 = \frac{1849}{6}$$

$$\Rightarrow 9933\lambda^2 - \frac{3311}{12} = 0 \Rightarrow \lambda^2 = \frac{1}{36} \Rightarrow \lambda = \pm\frac{1}{6}$$

When $\lambda = \frac{1}{6}$, $R\left(6+62\left(\frac{1}{6}\right), \frac{5}{2}+40\left(\frac{1}{6}\right), 4+67\left(\frac{1}{6}\right)\right) = R\left(\frac{49}{3}, \frac{55}{6}, \frac{91}{6}\right)$

When $\lambda = -\frac{1}{6}$, $R\left(6+62\left(-\frac{1}{6}\right), \frac{5}{2}+40\left(-\frac{1}{6}\right), 4+67\left(-\frac{1}{6}\right)\right) = R\left(-\frac{13}{3}, -\frac{25}{6}, -\frac{43}{6}\right)$

$\therefore R$ has coordinates $\left(\frac{49}{3}, \frac{55}{6}, \frac{91}{6}\right)$ or $\left(-\frac{13}{3}, -\frac{25}{6}, -\frac{43}{6}\right)$

12) a)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix}$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \\ -1 \end{pmatrix}$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 3 & -1 \\ -2 & 0 & -1 \end{vmatrix}$$

$$= (3 \times (-1) - (-1) \times 0)\mathbf{i} - \big(1 \times (-1) - (-1) \times (-2)\big)\mathbf{j} + \big(1 \times 0 - 3 \times (-2)\big)\mathbf{k}$$

$$= -3\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$$

So the plane has equation $-3x + 3y + 6z = d$

Substitute in the point $A(2, 1, 2)$ to find d:

$-3(2) + 3(1) + 6(2) = d \Rightarrow d = 9$

So the equation of the plane is $-3x + 3y + 6z = 9$ or $-x + y + 2z = 3$

12) b)

Midpoint of A and B is $\left(\frac{2+3}{2}, \frac{1+4}{2}, \frac{2+1}{2}\right) = \left(\frac{5}{2}, \frac{5}{2}, \frac{3}{2}\right)$

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix}$

The direction vector of the line, $\mathbf{d}$, must be perpendicular to the normal vector of the plane: $\mathbf{d} \cdot \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = 0$

And d must also be perpendicular to AB: $\mathbf{d} \cdot \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix} = 0$

Let $\mathbf{d} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$

$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = 0 \Rightarrow -a + b + 2c = 0$

$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix} = 0 \Rightarrow a + 3b - c = 0$

Let $c = 1$, then the equations become:

$-a + b + 2 = 0 \Rightarrow -a + b = -2$ (1)

$a + 3b - 1 = 0 \Rightarrow a + 3b = 1$ (2)

Solve (1) and (2) to find $a = \frac{7}{4}$ and $b = -\frac{1}{4}$

Then the equation of the line l_1 is $\mathbf{r} = \begin{pmatrix} \frac{5}{2} \\ \frac{5}{2} \\ \frac{3}{2} \end{pmatrix} + \lambda \begin{pmatrix} \frac{7}{4} \\ -\frac{1}{4} \\ 1 \end{pmatrix}$ or $\mathbf{r} = \begin{pmatrix} \frac{5}{2} \\ \frac{5}{2} \\ \frac{3}{2} \end{pmatrix} + \lambda \begin{pmatrix} 7 \\ -1 \\ 4 \end{pmatrix}$

12) c)

Midpoint of B and C is $\left(\frac{3+0}{2}, \frac{4+1}{2}, \frac{1+1}{2}\right) = \left(\frac{3}{2}, \frac{5}{2}, 1\right)$

$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ -3 \\ 0 \end{pmatrix}$

The direction vector of the line, $\mathbf{d}$, must be perpendicular to the normal vector of the plane: $\mathbf{d} \cdot \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = 0$

And d must also be perpendicular to AB: $\mathbf{d} \cdot \begin{pmatrix} -3 \\ -3 \\ 0 \end{pmatrix} = 0$

Let $\mathbf{d} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$

$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = 0 \Rightarrow -a + b + 2c = 0$

$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} -3 \\ -3 \\ 0 \end{pmatrix} = 0 \Rightarrow -3a - 3b = 0$

Let $c = 1$, then the equations become:

$-a + b + 2 = 0 \Rightarrow -a + b = -2$ (1)

$-3a + 3b = 0 \Rightarrow a + b = 0$ (2)

Solve (1) and (2) to find $a = 1$ and $b = 1$

Then the equation of the line l_2 is $\mathbf{r} = \begin{pmatrix} \frac{3}{2} \\ \frac{5}{2} \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$

12) d)

$l_1: \mathbf{r} = \begin{pmatrix} \frac{5}{2} \\ \frac{5}{2} \\ \frac{3}{2} \end{pmatrix} + \lambda \begin{pmatrix} 7 \\ -1 \\ 4 \end{pmatrix}$ and $l_2: \mathbf{r} = \begin{pmatrix} \frac{3}{2} \\ \frac{5}{2} \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$

$\frac{5}{2} + 7\lambda = \frac{3}{2} + \mu \Rightarrow 7\lambda - \mu = -1$ (1)

$\frac{5}{2} - \lambda = \frac{5}{2} - \mu \Rightarrow -\lambda + \mu = 0$ (2)

$\frac{3}{2} + 4\lambda = 1 + \mu \Rightarrow 4\lambda - \mu = -\frac{1}{2}$ (3)

Solve (1) and (2): $\lambda = -\frac{1}{6}$ and $\mu = -\frac{1}{6}$

Check in (3):

$4\left(-\frac{1}{6}\right) - \left(-\frac{1}{6}\right) = -\frac{1}{2}$ as required

So the lines intersect.

Substitute $\lambda = -\frac{1}{6}$ into l_1:

The lines intersect at $P\left(\frac{4}{3}, \frac{8}{3}, \frac{5}{6}\right)$

13) a)

Substitute $Q(10, k, 10)$ into the equation of the plane:

$(10) - (k) + 6(10) = 300 \Rightarrow 10 - k + 60 = 300 \Rightarrow k = -230$

13) b)

Let P be a point on the line, so $\overrightarrow{OP} = \begin{pmatrix} 50+5\lambda \\ -765+5\lambda \\ -116+\lambda \end{pmatrix}$

$\overrightarrow{QP} = \overrightarrow{OP} - \overrightarrow{OQ} = \begin{pmatrix} 50+5\lambda \\ -765+5\lambda \\ -116+\lambda \end{pmatrix} - \begin{pmatrix} 10 \\ -230 \\ 10 \end{pmatrix} = \begin{pmatrix} 40+5\lambda \\ -535+5\lambda \\ -126+\lambda \end{pmatrix}$

$\overrightarrow{QP} \cdot \begin{pmatrix} 5 \\ 5 \\ 1 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} 40+5\lambda \\ -535+5\lambda \\ -126+\lambda \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 5 \\ 1 \end{pmatrix} = 0$

$\Rightarrow (40+5\lambda) \times 5 + (-535+5\lambda) \times 5 + (-126+\lambda) \times 1 = 0$

$\Rightarrow 200 + 25\lambda - 2675 + 25\lambda - 126 + \lambda = 0$

$\Rightarrow 51\lambda - 2601 = 0$

$\lambda = 51$

So P has coordinates $\big(50+5(51), -765+5(51), -116+(51)\big) = (305, -510, -65)$

13) c)

So $\overrightarrow{QP} = \begin{pmatrix} 40+5(51) \\ -535+5(51) \\ -126+(51) \end{pmatrix} = \begin{pmatrix} 295 \\ -280 \\ -75 \end{pmatrix}$

So $\left|\overrightarrow{QP}\right| = \sqrt{295^2+(-280)^2+(-75)^2} = 5\sqrt{6842}$ units

Alternative Method (using the Vector Product):

Let $\mathbf{p} = \begin{pmatrix} 10 \\ -230 \\ 10 \end{pmatrix}$

Let A be a known point on the line so that its position vector is $\mathbf{a} = \begin{pmatrix} 50 \\ -765 \\ -116 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 5 \\ 5 \\ 1 \end{pmatrix}$

Then $h = \frac{|\mathbf{d}\times(\mathbf{p}-\mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix} 5 \\ 5 \\ 1 \end{pmatrix} \times \left(\begin{pmatrix} 10 \\ -230 \\ 10 \end{pmatrix} - \begin{pmatrix} 50 \\ -765 \\ -116 \end{pmatrix} \right)\right|}{\left|\begin{pmatrix} 5 \\ 5 \\ 1 \end{pmatrix}\right|} = \frac{\left|\begin{pmatrix} 5 \\ 5 \\ 1 \end{pmatrix} \times \begin{pmatrix} -40 \\ 535 \\ 126 \end{pmatrix}\right|}{\left|\begin{pmatrix} 5 \\ 5 \\ 1 \end{pmatrix}\right|}$

$$\begin{pmatrix} 5 \\ 5 \\ 1 \end{pmatrix} \times \begin{pmatrix} -40 \\ 535 \\ 126 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 5 & 5 & 1 \\ -40 & 535 & 126 \end{vmatrix}$$

$$= (5\times 126 - 1\times 535)\mathbf{i} - \big(5\times 126 - 1\times(-40)\big)\mathbf{j} + \big(5\times 535 - 5\times(-40)\big)\mathbf{k}$$

$$= 95\mathbf{i} - 670\mathbf{j} + 2875\mathbf{k}$$

$$\therefore h = \frac{\left|\begin{pmatrix} 95 \\ -670 \\ 2875 \end{pmatrix}\right|}{\left|\begin{pmatrix} 5 \\ 5 \\ 1 \end{pmatrix}\right|} = \frac{\sqrt{95^2+(-670)^2+2875^2}}{\sqrt{5^2+5^2+1^2}} = \frac{\sqrt{8723550}}{\sqrt{51}} = 5\sqrt{6842} \text{ units}$$

14) a)

Convert to column vectors: $\mathbf{r} = \begin{pmatrix}1\\1\\2\end{pmatrix} + \lambda\begin{pmatrix}1\\-1\\1\end{pmatrix}$

Let A be a point on the line, so $\overrightarrow{OA} = \begin{pmatrix}1+\lambda\\1-\lambda\\2+\lambda\end{pmatrix}$

$\overrightarrow{PA} = \overrightarrow{OA} - \overrightarrow{OP} = \begin{pmatrix}1+\lambda\\1-\lambda\\2+\lambda\end{pmatrix} - \begin{pmatrix}k\\2\\-1\end{pmatrix} = \begin{pmatrix}1-k+\lambda\\-1-\lambda\\3+\lambda\end{pmatrix}$

$\overrightarrow{PA} \cdot \begin{pmatrix}1\\-1\\1\end{pmatrix} = 0 \Rightarrow \begin{pmatrix}1-k+\lambda\\-1-\lambda\\3+\lambda\end{pmatrix} \cdot \begin{pmatrix}1\\-1\\1\end{pmatrix} = 0$

$\Rightarrow (1-k+\lambda)\times 1 + (-1-\lambda)\times(-1) + (3+\lambda)\times 1 = 0$

$\Rightarrow 1-k+\lambda+1+\lambda+3+\lambda = 0$

$\Rightarrow 3\lambda - k + 5 = 0$

$\lambda = \frac{k-5}{3}$

So $\overrightarrow{PA} = \begin{pmatrix}1-k+\left(\frac{k-5}{3}\right)\\-1-\left(\frac{k-5}{3}\right)\\3+\left(\frac{k-5}{3}\right)\end{pmatrix} = \begin{pmatrix}-\frac{2k+2}{3}\\\frac{2-k}{3}\\\frac{k+4}{3}\end{pmatrix}$

So $\left|\overrightarrow{PA}\right| = \sqrt{\left(-\frac{2k+2}{3}\right)^2 + \left(\frac{2-k}{3}\right)^2 + \left(\frac{k+4}{3}\right)^2} = \sqrt{\frac{1}{3^2}((2k+2)^2 + (2-k)^2 + (k+4)^2)}$

$= \frac{1}{3}\sqrt{4k^2+8k+4+k^2-4k+4+k^2+8k+16} = \frac{1}{3}\sqrt{6k^2+12k+24}$

Alternative Method (using the Vector Product):

Let $\mathbf{p} = \begin{pmatrix}k\\2\\-1\end{pmatrix}$

Let A be a known point on the line so that its position vector is $\mathbf{a} = \begin{pmatrix}1\\1\\2\end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix}1\\-1\\1\end{pmatrix}$

Then $h = \frac{|\mathbf{d}\times(\mathbf{p}-\mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix}1\\-1\\1\end{pmatrix}\times\left(\begin{pmatrix}k\\2\\-1\end{pmatrix}-\begin{pmatrix}1\\1\\2\end{pmatrix}\right)\right|}{\left|\begin{pmatrix}1\\-1\\1\end{pmatrix}\right|} = \frac{\left|\begin{pmatrix}1\\-1\\1\end{pmatrix}\times\begin{pmatrix}k-1\\1\\-3\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\-1\\1\end{pmatrix}\right|}$

$$\begin{pmatrix}1\\-1\\1\end{pmatrix}\times\begin{pmatrix}k-1\\1\\-3\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&-1&1\\k-1&1&-3\end{vmatrix}$$

$$=\big((-1)\times(-3)-1\times1\big)\mathbf{i}-\big(1\times(-3)-1\times(k-1)\big)\mathbf{j}+\big(1\times1-(-1)\times(k-1)\big)\mathbf{k}$$

$$=2\mathbf{i}+(k+2)\mathbf{j}+k\mathbf{k}$$

$$\therefore h=\frac{\left|\begin{pmatrix}2\\k+2\\k\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\-1\\1\end{pmatrix}\right|}=\frac{\sqrt{2^2+(k+2)^2+k^2}}{\sqrt{1^2+(-1)^2+1^2}}=\frac{\sqrt{2k^2+4k+8}}{\sqrt{3}}\times\frac{\sqrt{3}}{\sqrt{3}}=\frac{\sqrt{6k^2+12k+24}}{3}=\frac{1}{3}\sqrt{6k^2+12k+24}$$

14) b)

$$\frac{1}{3}\sqrt{6k^2+12k+24}=\frac{1}{3}\sqrt{6(k^2+2k)+24}=\frac{1}{3}\sqrt{6((k+1)^2-1)+24}$$

$$=\frac{1}{3}\sqrt{6(k+1)^2-6+24}=\frac{1}{3}\sqrt{6(k+1)^2+18}$$

Minimum value is $\frac{1}{3}\sqrt{18}=\sqrt{2}$ when $k=-1$

15)

Let P be a point on the line, so $\overrightarrow{OP} = \begin{pmatrix} 3\lambda \\ 1+\lambda \\ \lambda \end{pmatrix}$

$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} 3\lambda \\ 1+\lambda \\ \lambda \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1+3\lambda \\ -1+\lambda \\ -1+\lambda \end{pmatrix}$

$\overrightarrow{AP} \cdot \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} -1+3\lambda \\ -1+\lambda \\ -1+\lambda \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} = 0$

$\Rightarrow (-1+3\lambda) \times 3 + (-1+\lambda) \times 1 + (-1+\lambda) \times 1 = 0$

$\Rightarrow -3 + 9\lambda - 1 + \lambda - 1 + \lambda = 0$

$\Rightarrow 11\lambda - 5 = 0$

$\lambda = \frac{5}{11}$

So $\overrightarrow{AP} = \begin{pmatrix} -1+3\left(\frac{5}{11}\right) \\ -1+\left(\frac{5}{11}\right) \\ -1+\left(\frac{5}{11}\right) \end{pmatrix} = \begin{pmatrix} \frac{4}{11} \\ -\frac{6}{11} \\ -\frac{6}{11} \end{pmatrix}$

Let B be the image of A after being reflected in the line.

Then $\overrightarrow{OB} = \overrightarrow{OA} + 2\overrightarrow{AP} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + 2\begin{pmatrix} \frac{4}{11} \\ -\frac{6}{11} \\ -\frac{6}{11} \end{pmatrix} = \begin{pmatrix} \frac{19}{11} \\ \frac{10}{11} \\ -\frac{1}{11} \end{pmatrix}$

So B has coordinates $\left(\frac{19}{11}, \frac{10}{11}, -\frac{1}{11}\right)$

16)

Let P be a point on the line, so $\overrightarrow{OP} = \begin{pmatrix} 1+2\lambda \\ 4\lambda \\ 3-3\lambda \end{pmatrix}$

$$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} 1+2\lambda \\ 4\lambda \\ 3-3\lambda \end{pmatrix} - \begin{pmatrix} 29 \\ 29 \\ -\frac{109}{3} \end{pmatrix} = \begin{pmatrix} 2\lambda - 28 \\ 4\lambda - 29 \\ \frac{118}{3} - 3\lambda \end{pmatrix}$$

$$\overrightarrow{AP} \cdot \begin{pmatrix} 2 \\ 4 \\ -3 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} 2\lambda - 28 \\ 4\lambda - 29 \\ \frac{118}{3} - 3\lambda \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 4 \\ -3 \end{pmatrix} = 0$$

$$\Rightarrow (2\lambda - 28) \times 2 + (4\lambda - 29) \times 4 + \left(\frac{118}{3} - 3\lambda\right) \times (-3) = 0$$

$$\Rightarrow 4\lambda - 56 + 16\lambda - 116 - 118 + 9\lambda = 0$$

$$\Rightarrow 29\lambda - 290 = 0$$

$$\lambda = 10$$

So $\overrightarrow{AP} = \begin{pmatrix} 2(10) - 28 \\ 4(10) - 29 \\ \frac{118}{3} - 3(10) \end{pmatrix} = \begin{pmatrix} -8 \\ 11 \\ \frac{28}{3} \end{pmatrix}$

Let B be the image of A after being reflected in the line.

Then $\overrightarrow{OB} = \overrightarrow{OA} + 2\overrightarrow{AP} = \begin{pmatrix} 29 \\ 29 \\ -\frac{109}{3} \end{pmatrix} + 2\begin{pmatrix} -8 \\ 11 \\ \frac{28}{3} \end{pmatrix} = \begin{pmatrix} 13 \\ 51 \\ -\frac{53}{3} \end{pmatrix}$

So B has coordinates $\left(13, 51, -\frac{53}{3}\right)$

17)

Choose a point on l_1 so let A be the point with coordinates $(1, 0, 0)$

Let P be a point on the line l_2, so $\overrightarrow{OP} = \begin{pmatrix} 0 \\ \mu \\ 1+\mu \end{pmatrix}$

$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} 0 \\ \mu \\ 1+\mu \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ \mu \\ 1+\mu \end{pmatrix}$

$\overrightarrow{AP} \cdot \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} -1 \\ \mu \\ 1+\mu \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = 0$

$\Rightarrow -1 \times 0 + \mu \times 1 + (1+\mu) \times 1 = 0$

$\Rightarrow 0 + \mu + 1 + \mu = 0$

$\Rightarrow 2\mu + 1 = 0$

$\mu = -\frac{1}{2}$

So $\overrightarrow{AP} = \begin{pmatrix} -1 \\ -\frac{1}{2} \\ 1+\left(-\frac{1}{2}\right) \end{pmatrix} = \begin{pmatrix} -1 \\ -\frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$

So $\left|\overrightarrow{AP}\right| = \sqrt{(-1)^2 + \left(-\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{3}{2}} = \frac{\sqrt{6}}{2}$ units

Alternative Method (using the Vector Product to find the shortest distance between a point and a line):

Let $\mathbf{p} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

Let A be a known point on the line l_2 so that its position vector is $\mathbf{a} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$

Then $h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \times \left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right)\right|}{\left|\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}\right|} = \frac{\left|\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}\right|}{\left|\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}\right|}$

$\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{vmatrix}$

$= (1 \times (-1) - 1 \times 0)\mathbf{i} - (0 \times (-1) - 1 \times 1)\mathbf{j} + (0 \times 0 - 1 \times 1)\mathbf{k}$

$= -\mathbf{i} + \mathbf{j} - \mathbf{k}$

$\therefore h = \dfrac{\left|\begin{pmatrix}-1\\1\\-1\end{pmatrix}\right|}{\left|\begin{pmatrix}0\\1\\1\end{pmatrix}\right|} = \dfrac{\sqrt{(-1)^2+1^2+(-1)^2}}{\sqrt{0^2+1^2+1^2}} = \dfrac{\sqrt{3}}{\sqrt{2}} \times \dfrac{\sqrt{2}}{\sqrt{2}} = \dfrac{\sqrt{6}}{2}$ units

18)

Choose a point on l_1 so let A be the point with coordinates $(1, 0, 1)$

Let P be a point on the line l_2, so $\overrightarrow{OP} = \begin{pmatrix} 1+\mu \\ -1+\mu \\ 1+2\mu \end{pmatrix}$

$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} 1+\mu \\ -1+\mu \\ 1+2\mu \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \mu \\ -1+\mu \\ 2\mu \end{pmatrix}$

$\overrightarrow{AP} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} \mu \\ -1+\mu \\ 2\mu \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 0$

$\Rightarrow \mu \times 1 + (-1+\mu) \times 1 + 2\mu \times 2 = 0$

$\Rightarrow \mu - 1 + \mu + 4\mu = 0$

$\Rightarrow 6\mu - 1 = 0$

$\mu = \frac{1}{6}$

So $\overrightarrow{AP} = \begin{pmatrix} \left(\frac{1}{6}\right) \\ -1 + \left(\frac{1}{6}\right) \\ 2\left(\frac{1}{6}\right) \end{pmatrix} = \begin{pmatrix} \frac{1}{6} \\ -\frac{5}{6} \\ \frac{1}{3} \end{pmatrix}$

So $\left|\overrightarrow{AP}\right| = \sqrt{\left(\frac{1}{6}\right)^2 + \left(-\frac{5}{6}\right)^2 + \left(\frac{1}{3}\right)^2} = \frac{\sqrt{30}}{6}$ units

Alternative Method (using the Vector Product to find the shortest distance between a point and a line):

Let $\mathbf{p} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

Let A be a known point on the line l_2 so that its position vector is $\mathbf{a} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$

Then $h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \times \left(\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}\right)\right|}{\left|\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}\right|} = \frac{\left|\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right|}{\left|\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}\right|}$

$\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 2 \\ 0 & 1 & 0 \end{vmatrix}$

$= (1 \times 0 - 2 \times 1)\mathbf{i} - (1 \times 0 - 2 \times 0)\mathbf{j} + (1 \times 1 - 1 \times 0)\mathbf{k}$

$= -2\mathbf{i} + 0\mathbf{j} + \mathbf{k}$

$\therefore h = \frac{\left|\begin{pmatrix}-2\\0\\1\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\1\\2\end{pmatrix}\right|} = \frac{\sqrt{(-2)^2+0^2+1^2}}{\sqrt{1^2+1^2+2^2}} = \frac{\sqrt{5}}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = \frac{\sqrt{30}}{6}$ units

19)

Let $\mathbf{a}_1 = \begin{pmatrix}1\\0\\0\end{pmatrix}$ and $\mathbf{d}_1 = \begin{pmatrix}2\\1\\1\end{pmatrix}$

Let $\mathbf{a}_2 = \begin{pmatrix}0\\1\\0\end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix}1\\2\\2\end{pmatrix}$

Then $h = \left|\frac{\mathbf{d}_1 \times \mathbf{d}_2}{|\mathbf{d}_1 \times \mathbf{d}_2|} \cdot (\mathbf{a}_1 - \mathbf{a}_2)\right| = \left|\frac{\begin{pmatrix}2\\1\\1\end{pmatrix} \times \begin{pmatrix}1\\2\\2\end{pmatrix}}{\left|\begin{pmatrix}2\\1\\1\end{pmatrix} \times \begin{pmatrix}1\\2\\2\end{pmatrix}\right|} \cdot \left(\begin{pmatrix}1\\0\\0\end{pmatrix} - \begin{pmatrix}0\\1\\0\end{pmatrix}\right)\right|$

$\begin{pmatrix}2\\1\\1\end{pmatrix} \times \begin{pmatrix}1\\2\\2\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\2 & 1 & 1\\1 & 2 & 2\end{vmatrix}$

$= (1 \times 2 - 1 \times 2)\mathbf{i} - (2 \times 2 - 1 \times 1)\mathbf{j} + (2 \times 2 - 1 \times 1)\mathbf{k}$

$= 0\mathbf{i} - 3\mathbf{j} + 3\mathbf{k}$

$\therefore h = \left|\frac{\begin{pmatrix}0\\-3\\3\end{pmatrix}}{\left|\begin{pmatrix}0\\-3\\3\end{pmatrix}\right|} \cdot \begin{pmatrix}1\\-1\\0\end{pmatrix}\right| = \frac{\left|\begin{pmatrix}0\\-3\\3\end{pmatrix} \cdot \begin{pmatrix}1\\-1\\0\end{pmatrix}\right|}{\left|\begin{pmatrix}0\\-3\\3\end{pmatrix}\right|}$

$= \frac{|0\times1+(-3)\times(-1)+3\times0|}{\sqrt{0^2+(-3)^2+3^2}} = \frac{3}{\sqrt{18}} = \frac{\sqrt{2}}{2}$ units

20)

Let $\mathbf{a}_1 = \begin{pmatrix}20\\20\\60\end{pmatrix}$ and $\mathbf{d}_1 = \begin{pmatrix}80\\80\\90\end{pmatrix} - \begin{pmatrix}20\\20\\60\end{pmatrix} = \begin{pmatrix}60\\60\\30\end{pmatrix}$

Let $\mathbf{a}_2 = \begin{pmatrix}60\\20\\70\end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix}30\\70\\100\end{pmatrix} - \begin{pmatrix}60\\20\\70\end{pmatrix} = \begin{pmatrix}-30\\50\\30\end{pmatrix}$

Then $h = \left|\frac{\mathbf{d}_1\times\mathbf{d}_2}{|\mathbf{d}_1\times\mathbf{d}_2|}\cdot(\mathbf{a}_1-\mathbf{a}_2)\right| = \left|\frac{\begin{pmatrix}60\\60\\30\end{pmatrix}\times\begin{pmatrix}-30\\50\\30\end{pmatrix}}{\left|\begin{pmatrix}60\\60\\30\end{pmatrix}\times\begin{pmatrix}-30\\50\\30\end{pmatrix}\right|}\cdot\left(\begin{pmatrix}20\\20\\60\end{pmatrix}-\begin{pmatrix}60\\20\\70\end{pmatrix}\right)\right|$

$$\begin{pmatrix}60\\60\\30\end{pmatrix}\times\begin{pmatrix}-30\\50\\30\end{pmatrix} = \begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\60&60&30\\-30&50&30\end{vmatrix}$$

$$= (60\times30-30\times50)\mathbf{i} - \big(60\times30-30\times(-30)\big)\mathbf{j} + \big(60\times50-60\times(-30)\big)\mathbf{k}$$

$$= 300\mathbf{i} - 2700\mathbf{j} + 4800\mathbf{k}$$

$$\therefore h = \left|\frac{\begin{pmatrix}300\\-2700\\4800\end{pmatrix}}{\left|\begin{pmatrix}300\\-2700\\4800\end{pmatrix}\right|}\cdot\begin{pmatrix}-40\\0\\-10\end{pmatrix}\right| = \frac{\left|\begin{pmatrix}300\\-2700\\4800\end{pmatrix}\cdot\begin{pmatrix}-40\\0\\-10\end{pmatrix}\right|}{\left|\begin{pmatrix}300\\-2700\\4800\end{pmatrix}\right|}$$

$$= \frac{|300\times(-40)+(-2700)\times0+4800\times(-10)|}{\sqrt{300^2+(-2700)^2+4800^2}} = \frac{60000}{\sqrt{30420000}} = \frac{60000}{3900\sqrt{2}} = \frac{200}{13\sqrt{2}} = \frac{100\sqrt{2}}{13}\text{ m}$$

21)

First consider Drone 2 travelling from 3 metres above B to 4 metres above O:

Let $\mathbf{a}_1 = \begin{pmatrix} 0 \\ 9 \\ 18 \end{pmatrix}$ and $\mathbf{d}_1 = \begin{pmatrix} 15 \\ 6 \\ 4 \end{pmatrix} - \begin{pmatrix} 0 \\ 9 \\ 18 \end{pmatrix} = \begin{pmatrix} 15 \\ -3 \\ -14 \end{pmatrix}$

Let $\mathbf{a}_2 = \begin{pmatrix} 15 \\ 15 \\ 3 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix} - \begin{pmatrix} 15 \\ 15 \\ 3 \end{pmatrix} = \begin{pmatrix} -15 \\ -15 \\ 1 \end{pmatrix}$

Then $h = \left| \frac{\mathbf{d}_1 \times \mathbf{d}_2}{|\mathbf{d}_1 \times \mathbf{d}_2|} \cdot (\mathbf{a}_1 - \mathbf{a}_2) \right| = \left| \frac{\begin{pmatrix} 15 \\ -3 \\ -14 \end{pmatrix} \times \begin{pmatrix} -15 \\ -15 \\ 1 \end{pmatrix}}{\left| \begin{pmatrix} 15 \\ -3 \\ -14 \end{pmatrix} \times \begin{pmatrix} -15 \\ -15 \\ 1 \end{pmatrix} \right|} \cdot \left(\begin{pmatrix} 0 \\ 9 \\ 18 \end{pmatrix} - \begin{pmatrix} 15 \\ 15 \\ 3 \end{pmatrix} \right) \right|$

$$\begin{pmatrix} 15 \\ -3 \\ -14 \end{pmatrix} \times \begin{pmatrix} -15 \\ -15 \\ 1 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 15 & -3 & -14 \\ -15 & -15 & 1 \end{vmatrix}$$

$$= \big((-3) \times 1 - (-14) \times (-15)\big)\mathbf{i} - \big(15 \times 1 - (-14) \times (-15)\big)\mathbf{j} + \big(15 \times (-15) - (-3) \times (-15)\big)\mathbf{k}$$

$$= -213\mathbf{i} + 195\mathbf{j} - 270\mathbf{k}$$

$$\therefore h = \left| \frac{\begin{pmatrix} -213 \\ 195 \\ -270 \end{pmatrix}}{\left| \begin{pmatrix} -213 \\ 195 \\ -270 \end{pmatrix} \right|} \cdot \begin{pmatrix} -15 \\ -6 \\ 15 \end{pmatrix} \right| = \frac{\left| \begin{pmatrix} -213 \\ 195 \\ -270 \end{pmatrix} \cdot \begin{pmatrix} -15 \\ -6 \\ 15 \end{pmatrix} \right|}{\left| \begin{pmatrix} -213 \\ 195 \\ -270 \end{pmatrix} \right|}$$

$$= \frac{|(-213) \times (-15) + 195 \times (-6) + (-270) \times 15|}{\sqrt{(-213)^2 + 195^2 + (-270)^2}} = \frac{2025}{\sqrt{156294}} = \frac{2025}{3\sqrt{17366}} = \frac{675\sqrt{17366}}{17366} = 5.12 \text{ m (3 SF)}$$

Now consider Drone 2 travelling from 4 metres above O to 20 metres above C:

Let $\mathbf{a}_1 = \begin{pmatrix} 0 \\ 9 \\ 18 \end{pmatrix}$ and $\mathbf{d}_1 = \begin{pmatrix} 15 \\ 6 \\ 4 \end{pmatrix} - \begin{pmatrix} 0 \\ 9 \\ 18 \end{pmatrix} = \begin{pmatrix} 15 \\ -3 \\ -14 \end{pmatrix}$

Let $\mathbf{a}_2 = \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 15 \\ 12 \\ 20 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix} = \begin{pmatrix} 15 \\ 12 \\ 16 \end{pmatrix}$

Then $h = \left| \frac{\mathbf{d}_1 \times \mathbf{d}_2}{|\mathbf{d}_1 \times \mathbf{d}_2|} \cdot (\mathbf{a}_1 - \mathbf{a}_2) \right| = \left| \frac{\begin{pmatrix} 15 \\ -3 \\ -14 \end{pmatrix} \times \begin{pmatrix} 15 \\ 12 \\ 16 \end{pmatrix}}{\left| \begin{pmatrix} 15 \\ -3 \\ -14 \end{pmatrix} \times \begin{pmatrix} 15 \\ 12 \\ 16 \end{pmatrix} \right|} \cdot \left(\begin{pmatrix} 0 \\ 9 \\ 18 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix} \right) \right|$

$$\begin{pmatrix} 15 \\ -3 \\ -14 \end{pmatrix} \times \begin{pmatrix} 15 \\ 12 \\ 16 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 15 & -3 & -14 \\ 15 & 12 & 16 \end{vmatrix}$$

$$= \big((-3) \times 16 - (-14) \times 12\big)\mathbf{i} - (15 \times 16 - (-14) \times 15)\mathbf{j} + (15 \times 12 - (-3) \times 15)\mathbf{k}$$

$$= 120\mathbf{i} - 450\mathbf{j} + 225\mathbf{k}$$

$$\therefore h = \left|\frac{\begin{pmatrix}120\\-450\\225\end{pmatrix}}{\left|\begin{pmatrix}120\\-450\\225\end{pmatrix}\right|} \cdot \begin{pmatrix}0\\9\\14\end{pmatrix}\right| = \frac{\left|\begin{pmatrix}120\\-450\\225\end{pmatrix} \cdot \begin{pmatrix}0\\9\\14\end{pmatrix}\right|}{\left|\begin{pmatrix}120\\-450\\225\end{pmatrix}\right|}$$

$$= \frac{|120\times0+(-450)\times9+225\times14|}{\sqrt{120^2+(-450)^2+225^2}} = \frac{900}{\sqrt{267525}} = \frac{900}{15\sqrt{1189}} = \frac{60\sqrt{1189}}{1189} = 1.74 \text{ m (3 SF)}$$

So the flight path for Drone 2 is closer to the flight path for Drone 1 when it is travelling from O to C

22) a)

Let $\mathbf{a}_1 = \begin{pmatrix}-180\\-20\\25\end{pmatrix}$ and $\mathbf{d}_1 = \begin{pmatrix}260\\2\\36\end{pmatrix} - \begin{pmatrix}-180\\-20\\25\end{pmatrix} = \begin{pmatrix}440\\22\\11\end{pmatrix}$ or $\begin{pmatrix}40\\2\\1\end{pmatrix}$

Let $\mathbf{a}_2 = \begin{pmatrix}-220\\44\\-1\end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix}290\\10\\50\end{pmatrix} - \begin{pmatrix}-220\\44\\-1\end{pmatrix} = \begin{pmatrix}510\\-34\\51\end{pmatrix}$ or $\begin{pmatrix}30\\-2\\3\end{pmatrix}$

Then $h = \left|\frac{\mathbf{d}_1\times\mathbf{d}_2}{|\mathbf{d}_1\times\mathbf{d}_2|} \cdot (\mathbf{a}_1 - \mathbf{a}_2)\right| = \left|\frac{\begin{pmatrix}40\\2\\1\end{pmatrix}\times\begin{pmatrix}30\\-2\\3\end{pmatrix}}{\left|\begin{pmatrix}40\\2\\1\end{pmatrix}\times\begin{pmatrix}30\\-2\\3\end{pmatrix}\right|} \cdot \left(\begin{pmatrix}-180\\-20\\25\end{pmatrix} - \begin{pmatrix}-220\\44\\-1\end{pmatrix}\right)\right|$

$$\begin{pmatrix}40\\2\\1\end{pmatrix}\times\begin{pmatrix}30\\-2\\3\end{pmatrix} = \begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\40&2&1\\30&-2&3\end{vmatrix}$$

$$= \big(2\times3-1\times(-2)\big)\mathbf{i} - (40\times3-1\times30)\mathbf{j} + (40\times(-2)-2\times30)\mathbf{k}$$

$$= 8\mathbf{i} - 90\mathbf{j} - 140\mathbf{k}$$

$$\therefore h = \left|\frac{\begin{pmatrix}8\\-90\\-140\end{pmatrix}}{\left|\begin{pmatrix}8\\-90\\-140\end{pmatrix}\right|} \cdot \begin{pmatrix}40\\-64\\26\end{pmatrix}\right| = \frac{\left|\begin{pmatrix}8\\-90\\-140\end{pmatrix} \cdot \begin{pmatrix}40\\-64\\26\end{pmatrix}\right|}{\left|\begin{pmatrix}8\\-90\\-140\end{pmatrix}\right|}$$

$$= \frac{|8\times40+(-90)\times(-64)+(-140)\times26|}{\sqrt{8^2+(-90)^2+(-140)^2}} = \frac{2440}{\sqrt{27764}} = \frac{2440}{2\sqrt{6941}} = \frac{1220\sqrt{6941}}{6941} = 14.6 \text{ m (3 SF)}$$

$14.643\ldots > 10$ so the ski lifts pass the safety regulation.

22) b)

The ski-lifts are treated as particles without dimensions (they will likely be larger than half of $4.643\ldots$ m wide each, so it is likely the gap between the ski lifts will be shorter than 10 metres). We have also assumed that the ski lifts will travel along straight-line paths.

23)

Let $\mathbf{a}_1 = \begin{pmatrix}1\\0\\3\end{pmatrix}$ and $\mathbf{d}_1 = \begin{pmatrix}k\\2\\1\end{pmatrix}$

Let $\mathbf{a}_2 = \begin{pmatrix}3\\2\\1\end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix}4\\0\\1\end{pmatrix}$

Then $h = 2 \Rightarrow \left|\frac{\mathbf{d}_1\times\mathbf{d}_2}{|\mathbf{d}_1\times\mathbf{d}_2|}\cdot(\mathbf{a}_1-\mathbf{a}_2)\right| = 2 \Rightarrow \left|\frac{\begin{pmatrix}k\\2\\1\end{pmatrix}\times\begin{pmatrix}4\\0\\1\end{pmatrix}}{\left|\begin{pmatrix}k\\2\\1\end{pmatrix}\times\begin{pmatrix}4\\0\\1\end{pmatrix}\right|}\cdot\left(\begin{pmatrix}1\\0\\3\end{pmatrix}-\begin{pmatrix}3\\2\\1\end{pmatrix}\right)\right| = 2$

$$\begin{pmatrix}k\\2\\1\end{pmatrix}\times\begin{pmatrix}4\\0\\1\end{pmatrix} = \begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\k&2&1\\4&0&1\end{vmatrix}$$

$$= (2\times1-1\times0)\mathbf{i}-(k\times1-1\times4)\mathbf{j}+(k\times0-2\times4)\mathbf{k}$$

$$= 2\mathbf{i}+(4-k)\mathbf{j}-8\mathbf{k}$$

$$\therefore \left|\frac{\begin{pmatrix}2\\4-k\\-8\end{pmatrix}}{\left|\begin{pmatrix}2\\4-k\\-8\end{pmatrix}\right|}\cdot\begin{pmatrix}-2\\-2\\2\end{pmatrix}\right| = 2 \Rightarrow \frac{\left|\begin{pmatrix}2\\4-k\\-8\end{pmatrix}\cdot\begin{pmatrix}-2\\-2\\2\end{pmatrix}\right|}{\left|\begin{pmatrix}2\\4-k\\-8\end{pmatrix}\right|} = 2$$

$$\Rightarrow \frac{|2\times(-2)+(4-k)\times(-2)+(-8)\times2|}{\sqrt{2^2+(4-k)^2+(-8)^2}} = 2$$

$$\Rightarrow \frac{|2k-28|}{\sqrt{k^2-8k+84}} = 2$$

$$\Rightarrow |2k-28| = 2\sqrt{k^2-8k+84}$$

$$\Rightarrow (2k-28)^2 = 4(k^2-8k+84)$$

$$\Rightarrow 4k^2-112k+784 = 4k^2-32k+336$$

$$\Rightarrow 80k-448 = 0$$

$$\Rightarrow k = \frac{28}{5}$$

24)

Let $\mathbf{a}_1 = \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix}$ and $\mathbf{d}_1 = \begin{pmatrix} 3 \\ k \\ k \end{pmatrix}$

Let $\mathbf{a}_2 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$

Then $h = 1 \Rightarrow \left| \frac{\mathbf{d}_1 \times \mathbf{d}_2}{|\mathbf{d}_1 \times \mathbf{d}_2|} \cdot (\mathbf{a}_1 - \mathbf{a}_2) \right| = 1 \Rightarrow \left| \frac{\begin{pmatrix} 3 \\ k \\ k \end{pmatrix} \times \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}}{\left| \begin{pmatrix} 3 \\ k \\ k \end{pmatrix} \times \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right|} \cdot \left(\begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \right) \right| = 1$

$$\begin{pmatrix} 3 \\ k \\ k \end{pmatrix} \times \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & k & k \\ 1 & -1 & 0 \end{vmatrix}$$

$$= \big(k \times 0 - k \times (-1)\big)\mathbf{i} - (3 \times 0 - k \times 1)\mathbf{j} + (3 \times (-1) - k \times 1)\mathbf{k}$$

$$= k\mathbf{i} + k\mathbf{j} - (3 + k)\mathbf{k}$$

$$\therefore \left| \frac{\begin{pmatrix} k \\ k \\ -3-k \end{pmatrix}}{\left| \begin{pmatrix} k \\ k \\ -3-k \end{pmatrix} \right|} \cdot \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix} \right| = 1 \Rightarrow \frac{\left| \begin{pmatrix} k \\ k \\ -3-k \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix} \right|}{\left| \begin{pmatrix} k \\ k \\ -3-k \end{pmatrix} \right|} = 1$$

$$\Rightarrow \frac{|k \times (-1) + k \times (-1) + (-3-k) \times 0|}{\sqrt{k^2 + k^2 + (-3-k)^2}} = 1$$

$$\Rightarrow \frac{|-2k|}{\sqrt{3k^2 + 6k + 9}} = 1$$

$$\Rightarrow |-2k| = \sqrt{3k^2 + 6k + 9}$$

$$\Rightarrow (-2k)^2 = 3k^2 + 6k + 9$$

$$\Rightarrow 4k^2 = 3k^2 + 6k + 9$$

$$\Rightarrow k^2 - 6k - 9 = 0$$

$$\Rightarrow (k - 3)^2 - 9 - 9 = 0$$

$$\Rightarrow (k - 3)^2 = 18$$

$$\Rightarrow k = 3 \pm 3\sqrt{2}$$

25)

Let $\mathbf{a}_1 = \begin{pmatrix} -1 \\ k \\ 1 \end{pmatrix}$ and $\mathbf{d}_1 = \begin{pmatrix} k \\ 2 \\ -1 \end{pmatrix}$

Let $\mathbf{a}_2 = \begin{pmatrix} k \\ k \\ 1 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 3k \\ -2 \\ 2 \end{pmatrix}$

Then $h = 1 \Rightarrow \left| \frac{\mathbf{d}_1 \times \mathbf{d}_2}{|\mathbf{d}_1 \times \mathbf{d}_2|} \cdot (\mathbf{a}_1 - \mathbf{a}_2) \right| = 1 \Rightarrow \left| \frac{\begin{pmatrix} k \\ 2 \\ -1 \end{pmatrix} \times \begin{pmatrix} 3k \\ -2 \\ 2 \end{pmatrix}}{\left| \begin{pmatrix} k \\ 2 \\ -1 \end{pmatrix} \times \begin{pmatrix} 3k \\ -2 \\ 2 \end{pmatrix} \right|} \cdot \left(\begin{pmatrix} -1 \\ k \\ 1 \end{pmatrix} - \begin{pmatrix} k \\ k \\ 1 \end{pmatrix} \right) \right| = 1$

$$\begin{pmatrix} k \\ 2 \\ -1 \end{pmatrix} \times \begin{pmatrix} 3k \\ -2 \\ 2 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ k & 2 & -1 \\ 3k & -2 & 2 \end{vmatrix}$$

$$= \big(2 \times 2 - (-1) \times (-2)\big)\mathbf{i} - (k \times 2 - (-1) \times 3k)\mathbf{j} + (k \times (-2) - 2 \times 3k)\mathbf{k}$$

$$= 2\mathbf{i} - 5k\mathbf{j} - 8\mathbf{k}$$

$$\therefore \left| \frac{\begin{pmatrix} 2 \\ -5k \\ -8k \end{pmatrix}}{\left| \begin{pmatrix} 2 \\ -5k \\ -8k \end{pmatrix} \right|} \cdot \begin{pmatrix} -1-k \\ 0 \\ 0 \end{pmatrix} \right| = 1 \Rightarrow \frac{\left| \begin{pmatrix} 2 \\ -5k \\ -8k \end{pmatrix} \cdot \begin{pmatrix} -1-k \\ 0 \\ 0 \end{pmatrix} \right|}{\left| \begin{pmatrix} 2 \\ -5k \\ -8k \end{pmatrix} \right|} = 1$$

$$\Rightarrow \frac{|2 \times (-1-k) + (-5k) \times 0 + (-8k) \times 0|}{\sqrt{2^2 + (-5k)^2 + (-8k)^2}} = 1$$

$$\Rightarrow \frac{|-2k-2|}{\sqrt{89k^2+4}} = 1$$

$$\Rightarrow |-2k-2| = \sqrt{89k^2+4}$$

$$\Rightarrow (-2k-2)^2 = 89k^2 + 4$$

$$\Rightarrow 4k^2 + 8k + 4 = 89k^2 + 4$$

$$\Rightarrow 0 = 85k^2 - 8k$$

$$\Rightarrow 0 = (85k - 8)k$$

$\Rightarrow k = \frac{8}{85}$ or $k = 0$

As k is a positive constant, we must have $k = \frac{8}{85}$

26)

Let $\mathbf{a}_1 = \begin{pmatrix} 3.5 \\ k \\ 2.5 \end{pmatrix}$ and $\mathbf{d}_1 = \begin{pmatrix} 1.5 \\ 0.5 \\ 3 \end{pmatrix} - \begin{pmatrix} 3.5 \\ k \\ 2.5 \end{pmatrix} = \begin{pmatrix} -2 \\ 0.5-k \\ 0.5 \end{pmatrix}$

Let $\mathbf{a}_2 = \begin{pmatrix} 1.5 \\ 1 \\ 3 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 4 \\ 1.5 \\ 2.5 \end{pmatrix} - \begin{pmatrix} 1.5 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 2.5 \\ 0.5 \\ -0.5 \end{pmatrix}$

Then $h = 0.4 \Rightarrow \left| \frac{\mathbf{d}_1 \times \mathbf{d}_2}{|\mathbf{d}_1 \times \mathbf{d}_2|} \cdot (\mathbf{a}_1 - \mathbf{a}_2) \right| = 0.4 \Rightarrow \left| \frac{\begin{pmatrix} -2 \\ 0.5-k \\ 0.5 \end{pmatrix} \times \begin{pmatrix} 2.5 \\ 0.5 \\ -0.5 \end{pmatrix}}{\left| \begin{pmatrix} -2 \\ 0.5-k \\ 0.5 \end{pmatrix} \times \begin{pmatrix} 2.5 \\ 0.5 \\ -0.5 \end{pmatrix} \right|} \cdot \left(\begin{pmatrix} 3.5 \\ k \\ 2.5 \end{pmatrix} - \begin{pmatrix} 1.5 \\ 1 \\ 3 \end{pmatrix} \right) \right| = 0.4$

$$\begin{pmatrix} -2 \\ 0.5-k \\ 0.5 \end{pmatrix} \times \begin{pmatrix} 2.5 \\ 0.5 \\ -0.5 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 0.5-k & 0.5 \\ 2.5 & 0.5 & -0.5 \end{vmatrix}$$

$$= \big((0.5-k) \times (-0.5) - 0.5 \times 0.5\big)\mathbf{i} - \big((-2) \times (-0.5) - 0.5 \times 2.5\big)\mathbf{j} + \big((-2) \times 0.5 - (0.5-k) \times 2.5\big)\mathbf{k}$$

$$= \left(\frac{1}{2}k - \frac{1}{2}\right)\mathbf{i} + \frac{1}{4}\mathbf{j} + \left(\frac{5}{2}k - \frac{9}{4}\right)\mathbf{k}$$

$$\therefore \left| \frac{\begin{pmatrix} \frac{1}{2}k-\frac{1}{2} \\ \frac{1}{4} \\ \frac{5}{2}k-\frac{9}{4} \end{pmatrix}}{\left| \begin{pmatrix} \frac{1}{2}k-\frac{1}{2} \\ \frac{1}{4} \\ \frac{5}{2}k-\frac{9}{4} \end{pmatrix} \right|} \cdot \begin{pmatrix} 2 \\ k-1 \\ -\frac{1}{2} \end{pmatrix} \right| = 0.4 \Rightarrow \frac{\left| \begin{pmatrix} \frac{1}{2}k-\frac{1}{2} \\ \frac{1}{4} \\ \frac{5}{2}k-\frac{9}{4} \end{pmatrix} \cdot \begin{pmatrix} 2 \\ k-1 \\ -\frac{1}{2} \end{pmatrix} \right|}{\left| \begin{pmatrix} \frac{1}{2}k-\frac{1}{2} \\ \frac{1}{4} \\ \frac{5}{2}k-\frac{9}{4} \end{pmatrix} \right|} = 0.4$$

$$\Rightarrow \frac{\left|\left(\frac{1}{2}k-\frac{1}{2}\right)\times 2+\frac{1}{4}\times(k-1)+\left(\frac{5}{2}k-\frac{9}{4}\right)\times\left(-\frac{1}{2}\right)\right|}{\sqrt{\left(\frac{1}{2}k-\frac{1}{2}\right)^2+\left(\frac{1}{4}\right)^2+\left(\frac{5}{2}k-\frac{9}{4}\right)^2}} = 0.4$$

$$\Rightarrow \frac{\left|k-1+\frac{1}{4}k-\frac{1}{4}-\frac{5}{4}k+\frac{9}{8}\right|}{\sqrt{\frac{1}{4}k^2-\frac{1}{2}k+\frac{1}{4}+\frac{1}{16}+\frac{25}{4}k^2-\frac{45}{4}k+\frac{81}{16}}} = 0.4$$

$$\Rightarrow \frac{\left|-\frac{1}{8}\right|}{\sqrt{\frac{13}{2}k^2-\frac{47}{4}k+\frac{43}{8}}} = 0.4$$

$$\Rightarrow \frac{1}{8} = \frac{2}{5}\sqrt{\frac{13}{2}k^2 - \frac{47}{4}k + \frac{43}{8}}$$

$$\Rightarrow \frac{25}{256} = \frac{13}{2}k^2 - \frac{47}{4}k + \frac{43}{8}$$

$$\Rightarrow 0 = \frac{13}{2}k^2 - \frac{47}{4}k + \frac{1351}{256}$$

$$\Rightarrow 0 = k^2 - \frac{47}{26}k + \frac{1351}{1664}$$

$$\Rightarrow 0 = \left(k - \frac{47}{52}\right)^2 - \frac{2209}{2704} + \frac{1351}{1664}$$

$\Rightarrow \left(k - \frac{47}{52}\right)^2 = \frac{109}{21632}$

$\Rightarrow k = \frac{47}{52} \pm \frac{\sqrt{218}}{208}$

27)

Write $\frac{x+2}{4} = \frac{y-5}{2} = \frac{z+7}{3}$ as $l_1: \mathbf{r} = \begin{pmatrix} -2 \\ 5 \\ -7 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix}$

Write $\frac{x-1}{8} = \frac{y+2}{4} = \frac{z-4}{5}$ as $l_2: \mathbf{r} = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 8 \\ 4 \\ 5 \end{pmatrix}$

Let $\mathbf{a}_1 = \begin{pmatrix} -2 \\ 5 \\ -7 \end{pmatrix}$ and $\mathbf{d}_1 = \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix}$

Let $\mathbf{a}_2 = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 8 \\ 4 \\ 5 \end{pmatrix}$

Then $h = \left| \frac{\mathbf{d}_1 \times \mathbf{d}_2}{|\mathbf{d}_1 \times \mathbf{d}_2|} \cdot (\mathbf{a}_1 - \mathbf{a}_2) \right| = \left| \frac{\begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix} \times \begin{pmatrix} 8 \\ 4 \\ 5 \end{pmatrix}}{\left| \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix} \times \begin{pmatrix} 8 \\ 4 \\ 5 \end{pmatrix} \right|} \cdot \left(\begin{pmatrix} -2 \\ 5 \\ -7 \end{pmatrix} - \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix} \right) \right|$

$\begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix} \times \begin{pmatrix} 8 \\ 4 \\ 5 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 2 & 3 \\ 8 & 4 & 5 \end{vmatrix}$

$= (2 \times 5 - 3 \times 4)\mathbf{i} - (4 \times 5 - 3 \times 8)\mathbf{j} + (4 \times 4 - 2 \times 8)\mathbf{k}$

$= -2\mathbf{i} + 4\mathbf{j} + 0\mathbf{k}$

$\therefore h = \left| \frac{\begin{pmatrix} -2 \\ 4 \\ 0 \end{pmatrix}}{\left| \begin{pmatrix} -2 \\ 4 \\ 0 \end{pmatrix} \right|} \cdot \begin{pmatrix} -3 \\ 7 \\ -11 \end{pmatrix} \right| = \frac{\left| \begin{pmatrix} -2 \\ 4 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 7 \\ -11 \end{pmatrix} \right|}{\left| \begin{pmatrix} -2 \\ 4 \\ 0 \end{pmatrix} \right|}$

$= \frac{|(-2) \times (-3) + 4 \times 7 + 0 \times (-11)|}{\sqrt{(-2)^2 + 4^2 + 0^2}} = \frac{34}{\sqrt{20}} = \frac{17\sqrt{5}}{4} = 9.50$ units (3 SF)

28)

Write $x-1=\frac{y+3}{5}=\frac{z}{2}\Rightarrow\frac{x-1}{1}=\frac{y+3}{5}=\frac{z}{2}$ as $l_1:\mathbf{r}=\begin{pmatrix}1\\-3\\0\end{pmatrix}+\lambda\begin{pmatrix}1\\5\\2\end{pmatrix}$

Write $\frac{x+5}{2}=y-3=1-z\Rightarrow\frac{x+5}{2}=\frac{y-3}{1}=\frac{z-1}{-1}$ as $l_2:\mathbf{r}=\begin{pmatrix}-5\\3\\1\end{pmatrix}+\mu\begin{pmatrix}2\\1\\-1\end{pmatrix}$

Let $\mathbf{a}_1=\begin{pmatrix}1\\-3\\0\end{pmatrix}$ and $\mathbf{d}_1=\begin{pmatrix}1\\5\\2\end{pmatrix}$

Let $\mathbf{a}_2=\begin{pmatrix}-5\\3\\1\end{pmatrix}$ and $\mathbf{d}_2=\begin{pmatrix}2\\1\\-1\end{pmatrix}$

Then $h=\left|\frac{\mathbf{d}_1\times\mathbf{d}_2}{|\mathbf{d}_1\times\mathbf{d}_2|}\cdot(\mathbf{a}_1-\mathbf{a}_2)\right|=\left|\frac{\begin{pmatrix}1\\5\\2\end{pmatrix}\times\begin{pmatrix}2\\1\\-1\end{pmatrix}}{\left|\begin{pmatrix}1\\5\\2\end{pmatrix}\times\begin{pmatrix}2\\1\\-1\end{pmatrix}\right|}\cdot\left(\begin{pmatrix}1\\-3\\0\end{pmatrix}-\begin{pmatrix}-5\\3\\1\end{pmatrix}\right)\right|$

$$\begin{pmatrix}1\\5\\2\end{pmatrix}\times\begin{pmatrix}2\\1\\-1\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&5&2\\2&1&-1\end{vmatrix}$$

$$=(5\times(-1)-2\times1)\mathbf{i}-(1\times(-1)-2\times2)\mathbf{j}+(1\times1-5\times2)\mathbf{k}$$

$$=-7\mathbf{i}+5\mathbf{j}-9\mathbf{k}$$

$$\therefore h=\left|\frac{\begin{pmatrix}-7\\5\\-9\end{pmatrix}}{\left|\begin{pmatrix}-7\\5\\-9\end{pmatrix}\right|}\cdot\begin{pmatrix}6\\-6\\-1\end{pmatrix}\right|=\frac{\left|\begin{pmatrix}-7\\5\\-9\end{pmatrix}\cdot\begin{pmatrix}6\\-6\\-1\end{pmatrix}\right|}{\left|\begin{pmatrix}-7\\5\\-9\end{pmatrix}\right|}$$

$$=\frac{|(-7)\times6+5\times(-6)+(-9)\times(-1)|}{\sqrt{(-7)^2+5^2+(-9)^2}}=\frac{63}{\sqrt{155}}=\frac{63\sqrt{155}}{155}\text{ units}$$

29)

Write $2x+5=4-y=5-2z \Rightarrow \frac{x+\frac{5}{2}}{\frac{1}{2}}=\frac{y-4}{-1}=\frac{x-\frac{5}{2}}{-\frac{1}{2}}$ as $l_1: \mathbf{r}=\begin{pmatrix}-\frac{5}{2}\\4\\\frac{5}{2}\end{pmatrix}+\lambda\begin{pmatrix}\frac{1}{2}\\-1\\-\frac{1}{2}\end{pmatrix}$ or as

$l_1: \mathbf{r}=\begin{pmatrix}-\frac{5}{2}\\4\\\frac{5}{2}\end{pmatrix}+\lambda\begin{pmatrix}1\\-2\\-1\end{pmatrix}$

Write $\frac{x}{2}=4y, z=3 \Rightarrow \frac{x}{2}=\frac{y}{\frac{1}{4}}, z=3$ as $l_2: \mathbf{r}=\begin{pmatrix}0\\0\\3\end{pmatrix}+\mu\begin{pmatrix}2\\\frac{1}{4}\\0\end{pmatrix}$ or as $l_2: \mathbf{r}=\begin{pmatrix}0\\0\\3\end{pmatrix}+\mu\begin{pmatrix}8\\1\\0\end{pmatrix}$

Let $\mathbf{a}_1=\begin{pmatrix}-\frac{5}{2}\\4\\\frac{5}{2}\end{pmatrix}$ and $\mathbf{d}_1=\begin{pmatrix}1\\-2\\-1\end{pmatrix}$

Let $\mathbf{a}_2=\begin{pmatrix}0\\0\\3\end{pmatrix}$ and $\mathbf{d}_2=\begin{pmatrix}8\\1\\0\end{pmatrix}$

Then $h=\left|\frac{\mathbf{d}_1\times\mathbf{d}_2}{|\mathbf{d}_1\times\mathbf{d}_2|}\cdot(\mathbf{a}_1-\mathbf{a}_2)\right|=\left|\frac{\begin{pmatrix}1\\-2\\-1\end{pmatrix}\times\begin{pmatrix}8\\1\\0\end{pmatrix}}{\left|\begin{pmatrix}1\\-2\\-1\end{pmatrix}\times\begin{pmatrix}8\\1\\0\end{pmatrix}\right|}\cdot\left(\begin{pmatrix}-\frac{5}{2}\\4\\\frac{5}{2}\end{pmatrix}-\begin{pmatrix}0\\0\\3\end{pmatrix}\right)\right|$

$\begin{pmatrix}1\\-2\\-1\end{pmatrix}\times\begin{pmatrix}8\\1\\0\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&-2&-1\\8&1&0\end{vmatrix}$

$=\big((-2)\times0-(-1)\times1\big)\mathbf{i}-(1\times0-(-1)\times8)\mathbf{j}+(1\times1-(-2)\times8)\mathbf{k}$

$=\mathbf{i}-8\mathbf{j}+17\mathbf{k}$

$\therefore h=\left|\frac{\begin{pmatrix}1\\-8\\17\end{pmatrix}}{\left|\begin{pmatrix}1\\-8\\17\end{pmatrix}\right|}\cdot\begin{pmatrix}-\frac{5}{2}\\4\\-\frac{1}{2}\end{pmatrix}\right|=\frac{\left|\begin{pmatrix}1\\-8\\17\end{pmatrix}\cdot\begin{pmatrix}-\frac{5}{2}\\4\\-\frac{1}{2}\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\-8\\17\end{pmatrix}\right|}$

$=\frac{\left|1\times\left(-\frac{5}{2}\right)+(-8)\times4+17\times\left(-\frac{1}{2}\right)\right|}{\sqrt{1+(-8)^2+17^2}}=\frac{43}{\sqrt{354}}=\frac{43\sqrt{354}}{354}$ units

30)

Write $x-1=2-y=4-z \Rightarrow \frac{x-1}{1}=\frac{y-2}{-1}=\frac{z-4}{-1}$ as $l_1: \mathbf{r}=\begin{pmatrix}1\\2\\4\end{pmatrix}+\lambda\begin{pmatrix}1\\-1\\-1\end{pmatrix}$

Write $x=1,\ ky=z \Rightarrow x=1,\ \frac{y}{\frac{1}{k}}=\frac{z}{1}$ as $l_2: \mathbf{r}=\begin{pmatrix}1\\0\\0\end{pmatrix}+\mu\begin{pmatrix}0\\\frac{1}{k}\\1\end{pmatrix}$ or as $l_2: \mathbf{r}=\begin{pmatrix}1\\0\\0\end{pmatrix}+\mu\begin{pmatrix}0\\1\\k\end{pmatrix}$

Let $\mathbf{a}_1=\begin{pmatrix}1\\2\\4\end{pmatrix}$ and $\mathbf{d}_1=\begin{pmatrix}1\\-1\\-1\end{pmatrix}$

Let $\mathbf{a}_2=\begin{pmatrix}1\\0\\0\end{pmatrix}$ and $\mathbf{d}_2=\begin{pmatrix}0\\1\\k\end{pmatrix}$

Then $h=\sqrt{2} \Rightarrow \left|\frac{\mathbf{d}_1\times\mathbf{d}_2}{|\mathbf{d}_1\times\mathbf{d}_2|}\cdot(\mathbf{a}_1-\mathbf{a}_2)\right|=\sqrt{2} \Rightarrow \left|\frac{\begin{pmatrix}1\\-1\\-1\end{pmatrix}\times\begin{pmatrix}0\\1\\k\end{pmatrix}}{\left|\begin{pmatrix}1\\-1\\-1\end{pmatrix}\times\begin{pmatrix}0\\1\\k\end{pmatrix}\right|}\cdot\left(\begin{pmatrix}1\\2\\4\end{pmatrix}-\begin{pmatrix}1\\0\\0\end{pmatrix}\right)\right|=\sqrt{2}$

$$\begin{pmatrix}1\\-1\\-1\end{pmatrix}\times\begin{pmatrix}0\\1\\k\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&-1&-1\\0&1&k\end{vmatrix}$$

$$=\big((-1)\times k-(-1)\times 1\big)\mathbf{i}-(1\times k-(-1)\times 0)\mathbf{j}+(1\times 1-(-1)\times 0)\mathbf{k}$$

$$=(-k+1)\mathbf{i}-k\mathbf{j}+\mathbf{k}$$

$$\therefore \left|\frac{\begin{pmatrix}-k+1\\-k\\1\end{pmatrix}}{\left|\begin{pmatrix}-k+1\\-k\\1\end{pmatrix}\right|}\cdot\begin{pmatrix}0\\2\\4\end{pmatrix}\right|=\sqrt{2} \Rightarrow \frac{\left|\begin{pmatrix}-k+1\\-k\\1\end{pmatrix}\cdot\begin{pmatrix}0\\2\\4\end{pmatrix}\right|}{\left|\begin{pmatrix}-k+1\\-k\\1\end{pmatrix}\right|}=\sqrt{2}$$

$$\Rightarrow \frac{|(-k+1)\times 0+(-k)\times 2+1\times 4|}{\sqrt{(-k+1)^2+(-k)^2+1^2}}=\sqrt{2}$$

$$\Rightarrow \frac{|4-2k|}{\sqrt{2k^2-2k+2}}=\sqrt{2}$$

$$\Rightarrow |4-2k|=\sqrt{2}\sqrt{2k^2-2k+2}$$

$$\Rightarrow (4-2k)^2=2(2k^2-2k+2)$$

$$\Rightarrow 4k^2-16k+16=4k^2-4k+4$$

$$\Rightarrow -12k+12=0$$

$$\Rightarrow k=1$$

31) a)

The two lines are parallel as $\begin{pmatrix}-6\\5\\3\end{pmatrix}=-\begin{pmatrix}6\\-5\\-3\end{pmatrix}$, i.e. the direction vectors are multiples of one another.

31) b)

Let $l_1: \mathbf{r} = \begin{pmatrix} 7 \\ 6 \\ 10 \end{pmatrix} + \lambda \begin{pmatrix} -6 \\ 5 \\ 3 \end{pmatrix}$ and $l_2: \mathbf{r} = \begin{pmatrix} 0 \\ 2 \\ -9 \end{pmatrix} + \mu \begin{pmatrix} 6 \\ -5 \\ -3 \end{pmatrix}$

Choose a point on l_1 so let A be the point with coordinates $(7, 6, 10)$

Let P be a point on the line l_2, so $\overrightarrow{OP} = \begin{pmatrix} 6\mu \\ 2 - 5\mu \\ -9 - 3\mu \end{pmatrix}$

$$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} 6\mu \\ 2 - 5\mu \\ -9 - 3\mu \end{pmatrix} - \begin{pmatrix} 7 \\ 6 \\ 10 \end{pmatrix} = \begin{pmatrix} 6\mu - 7 \\ -4 - 5\mu \\ -19 - 3\mu \end{pmatrix}$$

$$\overrightarrow{AP} \cdot \begin{pmatrix} 6 \\ -5 \\ -3 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} 6\mu - 7 \\ -4 - 5\mu \\ -19 - 3\mu \end{pmatrix} \cdot \begin{pmatrix} 6 \\ -5 \\ -3 \end{pmatrix} = 0$$

$$\Rightarrow (6\mu - 7) \times 6 + (-4 - 5\mu) \times (-5) + (-19 - 3\mu) \times (-3) = 0$$

$$\Rightarrow 36\mu - 42 + 20 + 25\mu + 57 + 9\mu = 0$$

$$\Rightarrow 70\mu + 35 = 0$$

$$\mu = -\frac{1}{2}$$

So $\overrightarrow{AP} = \begin{pmatrix} 6\left(-\frac{1}{2}\right) - 7 \\ -4 - 5\left(-\frac{1}{2}\right) \\ -19 - 3\left(-\frac{1}{2}\right) \end{pmatrix} = \begin{pmatrix} -10 \\ -\frac{3}{2} \\ -\frac{35}{2} \end{pmatrix}$

So $\left|\overrightarrow{AP}\right| = \sqrt{(-10)^2 + \left(-\frac{3}{2}\right)^2 + \left(-\frac{35}{2}\right)^2} = \frac{\sqrt{1634}}{2}$ units

Alternative Method (using the Vector Product to find the shortest distance between a point and a line):

Let $\mathbf{p} = \begin{pmatrix} 7 \\ 6 \\ 10 \end{pmatrix}$

Let A be a known point on the line l_2 so that its position vector is $\mathbf{a} = \begin{pmatrix} 0 \\ 2 \\ -9 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 6 \\ -5 \\ -3 \end{pmatrix}$

Then $h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix} 6 \\ -5 \\ -3 \end{pmatrix} \times \left(\begin{pmatrix} 7 \\ 6 \\ 10 \end{pmatrix} - \begin{pmatrix} 0 \\ 2 \\ -9 \end{pmatrix}\right)\right|}{\left|\begin{pmatrix} 6 \\ -5 \\ -3 \end{pmatrix}\right|} = \frac{\left|\begin{pmatrix} 6 \\ -5 \\ -3 \end{pmatrix} \times \begin{pmatrix} 7 \\ 4 \\ 19 \end{pmatrix}\right|}{\left|\begin{pmatrix} 6 \\ -5 \\ -3 \end{pmatrix}\right|}$

$$\begin{pmatrix}6\\-5\\-3\end{pmatrix}\times\begin{pmatrix}7\\4\\19\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\6&-5&-3\\7&4&19\end{vmatrix}$$
$$=\big((-5)\times19-(-3)\times4\big)\mathbf{i}-(6\times19-(-3)\times7)\mathbf{j}+(6\times4-(-5)\times7)\mathbf{k}$$
$$=-83\mathbf{i}-135\mathbf{j}+59\mathbf{k}$$
$$\therefore h=\frac{\left|\begin{pmatrix}-83\\-135\\59\end{pmatrix}\right|}{\left|\begin{pmatrix}6\\-5\\-3\end{pmatrix}\right|}=\frac{\sqrt{(-83)^2+(-135)^2+59^2}}{\sqrt{6^2+(-5)^2+(-3)^2}}=\frac{\sqrt{28595}}{\sqrt{70}}=\frac{\sqrt{1634}}{2}\text{ units}$$

32)

Write $\left(\mathbf{r}-\begin{pmatrix}2\\1\\0\end{pmatrix}\right)\times\begin{pmatrix}0\\-1\\3\end{pmatrix}=\mathbf{0}$ as $l_1\colon\mathbf{r}=\begin{pmatrix}2\\1\\0\end{pmatrix}+\lambda\begin{pmatrix}0\\-1\\3\end{pmatrix}$

Write $\left(\mathbf{r}-\begin{pmatrix}-1\\5\\-3\end{pmatrix}\right)\times\begin{pmatrix}2\\5\\-1\end{pmatrix}=\mathbf{0}$ as $l_2\colon\mathbf{r}=\begin{pmatrix}-1\\5\\-3\end{pmatrix}+\mu\begin{pmatrix}2\\5\\-1\end{pmatrix}$

Let $\mathbf{a}_1=\begin{pmatrix}2\\1\\0\end{pmatrix}$ and $\mathbf{d}_1=\begin{pmatrix}0\\-1\\3\end{pmatrix}$

Let $\mathbf{a}_2=\begin{pmatrix}-1\\5\\-3\end{pmatrix}$ and $\mathbf{d}_2=\begin{pmatrix}2\\5\\-1\end{pmatrix}$

Then $h=\left|\frac{\mathbf{d}_1\times\mathbf{d}_2}{|\mathbf{d}_1\times\mathbf{d}_2|}\cdot(\mathbf{a}_1-\mathbf{a}_2)\right|=\left|\frac{\begin{pmatrix}0\\-1\\3\end{pmatrix}\times\begin{pmatrix}2\\5\\-1\end{pmatrix}}{\left|\begin{pmatrix}0\\-1\\3\end{pmatrix}\times\begin{pmatrix}2\\5\\-1\end{pmatrix}\right|}\cdot\left(\begin{pmatrix}2\\1\\0\end{pmatrix}-\begin{pmatrix}-1\\5\\-3\end{pmatrix}\right)\right|$

$$\begin{pmatrix}0\\-1\\3\end{pmatrix}\times\begin{pmatrix}2\\5\\-1\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\0&-1&3\\2&5&-1\end{vmatrix}$$
$$=\big((-1)\times(-1)-3\times5\big)\mathbf{i}-(0\times(-1)-3\times2)\mathbf{j}+(0\times5-(-1)\times2)\mathbf{k}$$
$$=-14\mathbf{i}+6\mathbf{j}+2\mathbf{k}$$
$$\therefore h=\left|\frac{\begin{pmatrix}-14\\6\\2\end{pmatrix}}{\left|\begin{pmatrix}-14\\6\\2\end{pmatrix}\right|}\cdot\begin{pmatrix}3\\-4\\3\end{pmatrix}\right|=\frac{\left|\begin{pmatrix}-14\\6\\2\end{pmatrix}\cdot\begin{pmatrix}3\\-4\\3\end{pmatrix}\right|}{\left|\begin{pmatrix}-14\\6\\2\end{pmatrix}\right|}$$
$$=\frac{|(-14)\times3+6\times(-4)+2\times3|}{\sqrt{(-14)^2+6^2+2^2}}=\frac{60}{\sqrt{236}}=\frac{30\sqrt{59}}{59}\text{ units}$$

33)

Write $2x+3=3-z, y=2 \Rightarrow \frac{x+\frac{3}{2}}{2}=\frac{z-3}{-1}, y=2$ as $l_1: \mathbf{r}=\begin{pmatrix}-\frac{3}{2}\\2\\3\end{pmatrix}+\lambda\begin{pmatrix}2\\0\\-1\end{pmatrix}$

Write $\left(\mathbf{r}-\begin{pmatrix}2\\0\\1\end{pmatrix}\right)\times\begin{pmatrix}-7\\2\\3\end{pmatrix}=\mathbf{0}$ as $l_2: \mathbf{r}=\begin{pmatrix}2\\0\\1\end{pmatrix}+\mu\begin{pmatrix}-7\\2\\3\end{pmatrix}$

Let $\mathbf{a}_1=\begin{pmatrix}-\frac{3}{2}\\2\\3\end{pmatrix}$ and $\mathbf{d}_1=\begin{pmatrix}2\\0\\-1\end{pmatrix}$

Let $\mathbf{a}_2=\begin{pmatrix}2\\0\\1\end{pmatrix}$ and $\mathbf{d}_2=\begin{pmatrix}-7\\2\\3\end{pmatrix}$

Then $h=\left|\frac{\mathbf{d}_1\times\mathbf{d}_2}{|\mathbf{d}_1\times\mathbf{d}_2|}\cdot(\mathbf{a}_1-\mathbf{a}_2)\right|=\left|\frac{\begin{pmatrix}2\\0\\-1\end{pmatrix}\times\begin{pmatrix}-7\\2\\3\end{pmatrix}}{\left|\begin{pmatrix}2\\0\\-1\end{pmatrix}\times\begin{pmatrix}-7\\2\\3\end{pmatrix}\right|}\cdot\left(\begin{pmatrix}-\frac{3}{2}\\2\\3\end{pmatrix}-\begin{pmatrix}2\\0\\1\end{pmatrix}\right)\right|$

$$\begin{pmatrix}2\\0\\-1\end{pmatrix}\times\begin{pmatrix}-7\\2\\3\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&0&-1\\-7&2&3\end{vmatrix}$$

$$=(0\times3-(-1)\times2)\mathbf{i}-\big(2\times3-(-1)\times(-7)\big)\mathbf{j}+\big(2\times2-0\times(-7)\big)\mathbf{k}$$

$$=2\mathbf{i}+\mathbf{j}+4\mathbf{k}$$

$$\therefore h=\left|\frac{\begin{pmatrix}2\\1\\4\end{pmatrix}}{\left|\begin{pmatrix}2\\1\\4\end{pmatrix}\right|}\cdot\begin{pmatrix}-\frac{7}{2}\\2\\2\end{pmatrix}\right|=\frac{\left|\begin{pmatrix}2\\1\\4\end{pmatrix}\cdot\begin{pmatrix}-\frac{7}{2}\\2\\2\end{pmatrix}\right|}{\left|\begin{pmatrix}2\\1\\4\end{pmatrix}\right|}$$

$$=\frac{\left|2\times\left(-\frac{7}{2}\right)+1\times2+4\times2\right|}{\sqrt{2^2+1^2+4^2}}=\frac{3}{\sqrt{21}}=\frac{\sqrt{21}}{7}\text{ units}$$